

AP Computer Science Principles

Guided Notes · Units 1–5



How to use these notes

Each topic has two pages. **Page 1** covers the core ideas and vocabulary — fill in every blank. **Page 2** goes deeper and ends with exam-style practice that gets harder as you go:

★ Warm-up

★★ On Level

★★★ Exam Hard

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Reference sheet reminder:

All code uses the AP exam reference sheet pseudocode. Lists start at index 1, ← means assignment, and `RANDOM(a, b)` includes both a and b.

Unit 1.2 Program Function and Purpose



The Big Picture

Every program is built for a reason. On the exam (and in your Create Performance Task) you must describe **why** a program exists, **what** it does, and identify its **inputs** and **outputs**.

Purpose vs. Function

- **Purpose** = _____ the program exists: the problem it solves or the creative interest it pursues.
- **Function** = _____ the program does when it runs (its behavior).
- Knowing the purpose helps developers build a program that actually meets users' _____.

Inputs → Program → Outputs

- **Input** = data _____ a computer for processing. It can come from a user, a _____, or another program.
- **Output** = data _____ a program to a device. It can be text, visuals, audio, or _____ (touch/vibration).
- Inputs usually _____ the output a program produces.

Describe the behavior

You don't need to read every line of code to describe a program. Describe what happens from the user's point of view:

"When the user _____, the program _____."



Think:

Purpose answers WHY? **Function** answers WHAT does it do?

Input/Output answer WHAT goes in and WHAT comes out?

Key Vocabulary

a collection of _____

Program

that performs a specific task when run by a computer (also called software)

Code segment

a collection of program statements that is _____ a program

Behavior

how a program functions during _____; often described by how a user interacts with it

Event

an action that supplies _____ data to a program (click, key press, timer)

Event-driven programming

program statements run when _____ by an event, not in a set order

You Try!

A fitness app lets users type the minutes they exercised each day. When the user taps **Weekly Report**, it shows a bar graph and plays a cheer sound if they reached 150 minutes.

(a) State the app's purpose. (b) Identify two inputs and two outputs. (c) Which input is an event?

Unit 1.2 Program Function and Purpose

Go Deeper & Exam Practice

Events in Action

```
ON buttonClick("submit")
{
  score ← score + 1
  DISPLAY(score)
}
```

The event is _____. The two lines inside only run _____, not when the program first starts. Clicking 3 times displays _____ (if score starts at 0).

Kinds of Input & Output

Input sources	Output types
keyboard, mouse, touchscreen	text on a screen
microphone, camera	images / video
GPS, temperature _____	sound, speech
data from another program/file	_____ (tactile)

A smartwatch heart-rate reading is an _____.
A game controller rumbling is _____ output.

Exam Practice

★ Warm-up

1. Which best describes the *purpose* of a program?
- The programming language it is written in
 - The number of lines of code it contains
 - The problem it solves or the creative interest it pursues
 - The hardware it runs on

★★ On Level

2. A smart thermostat reads the room temperature every minute and turns on the heater if it drops below a setting. Which is an **input**?
- The heater turning on
 - The electricity used by the heater
 - The light on the thermostat's screen
 - The room temperature reading

★★ On Level

3. Which is the best example of **event-driven** behavior?
- A program plays a sound when the user presses the spacebar
 - A program computes the average of a fixed list and stops
 - A program prints the numbers 1 to 10
 - A program sorts a list of names alphabetically

★★★ Exam Hard

4. A weather app is described: "When the user enters a city name, it displays the current temperature and a weather icon." Which TWO statements are true? (*Select two.*)
- The city name is an input.
 - The weather icon is an output.
 - The app's purpose is to display an icon.
 - Displaying the temperature is an event.

Challenge!

Pick an app you use daily. In complete sentences, describe its **purpose**, one of its **functions**, one **input** that is an **event**, and one **output** — exactly the way you would on the Create Performance Task written response.

Unit 1.3 Program Design and Development



The Big Picture

Programs aren't written in one try. Developers investigate, design, prototype, and test — over and over — while documenting their work. Know **iterative** vs. **incremental** development and why documentation and credit matter.

The Development Process

A development process can be ordered and intentional, or exploratory. Common phases:

Investigate → _____ → **Prototype** → _____

- Investigation identifies program _____ using surveys, interviews, user testing, and _____.

The Design Phase

Design can include:

- brainstorming
- planning and _____
- organizing the program into _____ and functional components
- diagrams of the user _____ layout
- developing a _____ strategy

Document & Give Credit

- Documentation helps in developing and _____ correct programs, especially when working with others.
- Document _____ development, not just at the end.
- When you use code written by someone else (or from a library), you must _____ it by naming the source or original author.



Think:

Iterative = loop back and improve using feedback. **Incremental** = build one working piece at a time. A real project is usually BOTH!

Key Vocabulary

Iterative development

requires _____

based on feedback, testing, or reflection; may mean revisiting earlier phases

Incremental development

breaks the problem into _____ and makes sure each piece works before adding it to the whole

Program requirements

describe how a program _____; may include descriptions of user interactions

Program documentation

a written description of the _____ of a code segment, event, procedure, or program and how it was developed

Comments

documentation written _____ that is not executed

You Try!

A team builds a cafeteria-ordering app. They first get the **menu screen** fully working, then add the **cart**, then **payment**. After a beta test, students say the buttons are too small, so the team goes back and redesigns the screens. Which part shows **incremental** development? Which part shows **iterative** development? Explain.

Unit 1.3 Program Design and Development

Go Deeper & Exam Practice

Investigate Your Users

Ways to gather requirements: _____, _____, user testing, and direct observation.

Why bother? A program can match its specification but still fail users if developers never asked about their _____. Investigating early means _____ costly changes later.

Testing with a _____ group of users helps catch problems the team didn't think of.

Comments in Code

```
# Calculates a 15% tip on the bill
# Rounding idea adapted from a
# CS Forum post by user "jkim"
tip ← bill * 0.15
```

- Comments do not affect how a program _____.
- Not all programming environments support comments, so other forms of _____ may be needed.
- The second comment gives _____ for borrowed ideas.

Exam Practice

★ Warm-up

1. Which is the best example of program documentation?
- A comment explaining what a procedure does and its parameters
 - A variable named x
 - A faster sorting algorithm
 - Removing unused code

★★ On Level

2. Testers report a confusing menu. The developer returns to the design phase to change it before continuing. This is an example of...
- iterative development
 - incremental development
 - a syntax error
 - procedural abstraction

★★ On Level

3. Which is a primary reason programmers write comments?
- To make the program run faster
 - To help people understand and maintain the code
 - To fix syntax errors automatically
 - To reduce the size of the program file

★★★ Exam Hard

4. A student uses a sorting procedure found in an online tutorial in their Create task. What is the most appropriate action?
- Use it without mention, since it is public
 - Use it and cite the source/original author in the program and documentation
 - Rename its variables so it counts as original
 - Never use outside code under any circumstances

Challenge!

Explain how investigating users *early* (surveys, interviews, observation) can reduce the amount of iterative revision needed *later*. Then describe one thing that investigation might still miss.

Unit 1.4 Identifying and Correcting Errors



The Big Picture

Every programmer makes mistakes. Know the **four types of errors**, the strategies used to find them, and how to pick test inputs. The exam LOVES asking “which input reveals the bug?”

Find & Fix Strategies

- writing _____
- hand _____ (tracking variable values on paper)
- visualizations
- _____ (step through code line by line)
- adding extra _____ to see values while the program runs

Testing with Inputs

- Testing uses defined inputs to check that an algorithm produces the _____.
- Program _____ tell you which inputs to test.
- Test inputs at and just beyond the _____ (minimum and maximum) of the input data — these are called **boundary / edge cases**.

Name That Error

Code	Error type
DISPLAY("Hi" (missing parenthesis)	_____
$x \leftarrow 10 / 0$	_____
$\text{avg} \leftarrow a + b / 2$	_____



Think:

Syntax = won't even run. Run-time = crashes WHILE running. Logic = runs fine, WRONG answer. Overflow = number too big for its space.

Key Vocabulary

Logic error

a mistake in the algorithm that causes _____

behavior; the program runs but is wrong

Syntax error

the rules of the programming _____ are not followed

Run-time error

an error that occurs _____ (e.g., dividing by zero, a bad list index)

Overflow error

a computer tries to handle a number outside its defined _____

Debugging

the process of finding and _____ errors

You Try!

This procedure should return the largest value in a list, but sometimes it's wrong.

```
PROCEDURE findMax(nums)
{
  max ← 0
  FOR EACH n IN nums
  {
    IF (n > max) { max ← n }
  }
  RETURN (max)
}
```

What type of error is this? Give a list that reveals it, and fix the code.

Date:

Go Deeper

A teen-club sign-up app should accept ages **13 through 18**. Explain why testing only “typical” ages like 15 is not enough, then list four test inputs you would use and the expected result for each.

Unit 1.4 Identifying and Correcting Errors

Exam Practice

Exam Practice

★ Warm-up

1. A program runs without crashing but displays the wrong average. What type of error is most likely?

- A. Syntax error
- B. Run-time error
- C. Logic error
- D. Overflow error

★★ On Level

2. In a language that stores integers using 8 bits (0 to 255), a program adds 1 to 255. What kind of error can occur?

- A. Syntax error
- B. Overflow error
- C. Logic error
- D. Roundoff error

★★ On Level

3. Which strategy is LEAST helpful for finding a logic error?

- A. Hand tracing with a trace table
- B. Adding DISPLAY statements to show variable values
- C. Making all variable names shorter
- D. Using a debugger to step through code

★★★ Exam Hard

4. This code should count values **greater than 10**:

```
count ← 0
FOR EACH val IN myList
{
  IF (val ≥ 10)
  { count ← count + 1 }
}
```

Which test list reveals the error?

- A. [10, 4, 2]
- B. [12, 15, 20]
- C. [5, 8, 3]
- D. []

Show Your Work

Unit 2.1 Binary Numbers



The Big Picture

Everything a computer stores — numbers, text, images, sound — is ultimately **bits**. You must convert between binary and decimal, know how many values n bits can represent, and explain overflow and roundoff errors.

Bits & Bytes

- A **bit** (binary digit) is either _____.
- A **byte** is _____ bits.
- The same sequence of bits can represent different types of data (number, color, letter) in different _____.
- With n bits you can represent _____ different values.

Place Values

Decimal is base _____. Binary is base _____ — each place is worth double the one to its right.

128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	1

00001011 = $8 + 2 + 1 =$ _____ · 25 in binary = _____

Limits of Fixed Bits

- In many languages, integers use a _____ number of bits, which limits the range → overflow.
- Some languages (like Python) let integers grow as large as computer _____ allows.
- Real numbers are approximated, so $0.1 + 0.2$ may show 0.30000000000000004 — a _____ error.



Think:

Each extra bit **DOUBLES** the number of values: 3 bits → 8 values, 4 bits → 16, 8 bits → 256. The largest value with n bits is $2^n - 1$.

Key Vocabulary

Bit

short for binary digit; the _____ unit of data, 0 or 1

Analog data

data with values that change _____ over time (pitch, volume, position)

Sampling

approximating analog data digitally by measuring values at _____

Overflow error

occurs when a number is too _____ for the fixed number of bits used to store it

Roundoff error

real numbers are _____ because only a fixed number of bits is used

You Try!

- Convert 10110 to decimal.
- Convert 45 to binary.
- A school needs a unique binary ID for each of its 300 students. What is the minimum number of bits needed? Explain.

Unit 2.1 Binary Numbers

Go Deeper & Exam Practice

Decimal → Binary (subtract method)

Convert **77**: start at the biggest place value that fits.

- 64 fits → write 1, remainder _____
 - 32? no → 0 16? no → 0
 - 8 fits → 1, remainder _____
 - 4 fits → 1, remainder _____
 - 2? no → 0 1 fits → 1
- 77 = _____ (7 bits)

Doubling Table

Bits	Values	Largest number
1	2	1
3	_____	_____
4	_____	_____
8	_____	_____
n + 1	_____ values as n bits	

Sampling more often makes digital audio _____ but uses more bits.

Exam Practice

★ Warm-up

1. What is the decimal value of binary **1101**?
- A. 13
B. 11
C. 14
D. 1101

★★ On Level

2. Which list of binary numbers is in order from **least to greatest**?
- A. 1001, 0111, 1100, 1010
B. 0111, 1010, 1001, 1100
C. 1100, 1010, 1001, 0110
D. 0110, 1001, 1010, 1100

★★ On Level

3. A school stores locker numbers using 6 bits. It now needs 100 lockers. Which is true?
- A. 6 bits represents only 64 values; at least 7 bits are needed
B. 6 bits is enough because $6 \times 20 = 120$
C. 100 bits are needed, one per locker
D. 8 bits are the minimum because a byte is 8 bits

★★★ Exam Hard

4. A game stores a player's score using **4 bits** (values 0–15). The player has 14 points and earns 3 more. What is most likely to happen?
- A. The score correctly shows 17
B. An overflow error occurs
C. A roundoff error occurs
D. A syntax error occurs

Challenge!

IPv4 addresses use **32 bits**. IPv6 addresses use **128 bits**. How many times more addresses does IPv6 allow than IPv4? Show your reasoning with powers of 2, and explain why this change was needed.

Unit 2.2 Data Compression



The Big Picture

Compression reduces the number of bits needed to store or send data. Know **lossless vs. lossy**, when to choose each, and how to apply a simple scheme like run-length encoding.

Lossless

- Reduces bits while guaranteeing _____ of the original data.
- Use it when you need the _____ original back: text, code, spreadsheets, medical images.
- Examples: run-length encoding, ZIP, PNG.

Lossy

- Can greatly reduce bits but only reconstructs an _____ of the original.
- Usually achieves _____ reduction than lossless.
- Use it when minimizing size or _____ matters most: streaming video, photos, music (JPEG, MP3).

Key Vocabulary

Data compression

reducing the number of _____ needed to represent data

Lossless

compression where _____ is lost; the original can be restored

Lossy

compression that _____

some data

Run-length encoding

replaces runs of repeated values with a _____ and the value

Redundancy

_____ patterns in data that compression takes advantage of

What Affects Compression?

How much a file shrinks depends on:

- the amount of _____ in the original data, and
- the _____ applied.

Trade-off: lossy = smaller but lower _____; lossless = exact but bigger.



Think:

Ask: "Do I need it back EXACTLY?"
Yes → lossless. "Is small and fast more important than perfect?" → lossy.

You Try!

Use run-length encoding to compress:

AAAAABBBCCCCCCCCD

(a) Compressed string? (b) Characters before vs. after? (c) Why would RLE make ABCDEFG *bigger*?

Unit 2.2 Data Compression

Go Deeper & Exam Practice

RLE on Images

One row of pixels (W = white, B = black):

W W W W B B W W W W W W

Encoded: _____

Decode 3B 2W 1R: _____

Which compresses better with RLE: a cartoon with big solid areas or a photo of a forest?

_____ — more _____.

Pattern Substitution

Replace every at with *:

the cat sat on the mat (22 chars)

→ the c* s* on the m* (_____ chars)

The program must also store the key * = at

so the text can be _____.

Is this lossy or lossless? _____, because the original can be rebuilt exactly.

Exam Practice

★ Warm-up

1. Which situation most requires **lossless** compression?
- A. Streaming a movie
 - B. A video call with friends
 - C. Posting a profile picture
 - D. Sending a spreadsheet of student grades

★★ On Level

2. Which statement about lossy compression is true?
- A. The original can always be perfectly restored
 - B. It only works on text files
 - C. It usually achieves a greater reduction in size than lossless
 - D. It never changes the quality of an image

★★ On Level

3. A photo app wants to minimize upload time and storage. Users don't notice small quality loss. Which is the best choice?
- A. Lossy, because size and speed matter more than perfect quality
 - B. Lossless, because it keeps every detail
 - C. No compression
 - D. Lossless, because it always makes smaller files

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
- A. Lossless algorithms can guarantee reconstruction of the original data.
 - B. Lossy compression always produces a smaller file than lossless for every file.
 - C. The amount of compression depends on the redundancy in the data.
 - D. Compressing a file twice with a lossless algorithm always makes it smaller again.

Challenge!

A hospital wants to save storage space on its X-ray images. An administrator suggests **lossy** compression because it saves the most space. Argue for or against this plan, explaining the trade-off.

Unit 2.3 Extracting Information from Data



The Big Picture

Data alone isn't information. Know how people and programs find patterns, what **metadata** is, what makes large data sets challenging, and why **correlation does not mean causation**.

Data → Information

- Data provide opportunities for identifying _____, making connections, and addressing problems.
- Digitally processed data may show a correlation between variables, but a correlation does not necessarily indicate _____.
- Patterns can appear when data are transformed using _____.

Metadata

- Metadata are data _____.
- Example: a photo's date, time, _____, and file size.
- Changing or deleting metadata does not change the _____.
- Metadata help us find, organize, and _____ information.

Challenges with Data

- Data may be _____ or invalid (missing values, wrong formats) → needs cleaning.
- Large data sets may need _____ systems; they're too big for one computer.
- Data sets can be biased, so conclusions may not apply to _____.



Think:

Ice cream sales and sunburns both rise in summer. Does ice cream CAUSE sunburn? No! A third factor — hot, sunny weather — drives both.

Key Vocabulary

Information

the collection of _____
extracted from data

Correlation

a _____
between two variables;
one changes as the other
changes

Data cleaning

making data _____
without changing its
meaning (e.g., fixing
spellings/abbreviations)

Scalability

a system's capacity to
change in _____
to
meet new demands

Bias in data

caused by the type or
_____ of data; NOT
fixed by just collecting
more data

You Try!

A survey's "Grade" column contains: 9, 9th, ninth, 10, Tenth, and blank cells.

(a) What process should be done first? (b) What problem do blanks cause? (c) The survey shows students with more screen time have lower grades. Can we conclude screen time *causes* lower grades?

Unit 2.3 Extracting Information from Data

Go Deeper & Exam Practice

Metadata in the Real World

Data	Example metadata
Photo	date/time, GPS _____, camera model
Text message	sender, _____, recipient
Song file	artist, _____, genre

Even without reading your messages, metadata can reveal who you talk to and _____ — a _____ concern (see 5.6).

Tools for Finding Patterns

- Combining data sources, _____ data, and classifying data are parts of processing information.
- _____ like charts and graphs make trends easy to see.
- Programs can filter and sort millions of rows that people could never handle _____.
- Insight comes from translating and _____ digital information.

Exam Practice

★ Warm-up

1. Which is an example of **metadata** for a digital photo?

- A. The colors of the pixels
- B. The date the photo was taken
- C. The people in the photo
- D. The brightness of the image

★★ On Level

2. A data set lists states as “CA”, “Calif.”, and “California”. Which process should be applied?

- A. Data compression
- B. Encryption
- C. Data cleaning
- D. Sampling

★★ On Level

3. Researchers find that towns with more libraries tend to have more crimes. Which conclusion is most appropriate?

- A. Libraries cause crime
- B. The variables are correlated; another factor (like population) may explain both
- C. Closing libraries would lower crime
- D. The data must be wrong

★★★ Exam Hard

4. A music service stores for each song: title, artist, genre, length, and number of plays. Which question **CANNOT** be answered using only this data?

- A. Which genre has the most total plays?
- B. What is the average song length for each artist?
- C. Which age group listens to the most songs?
- D. Which artist has the most songs in the library?

Challenge!

A company tests a voice assistant using recordings from 10,000 employees at its headquarters. Engineers suggest collecting 50,000 more recordings from the *same* office to reduce bias. Explain why this will not fix the problem and what they should do instead.

Unit 2.4 Using Programs with Data



The Big Picture

Programs let us process data sets far too large to handle by hand. Know what programs do with data (filter, sort, combine, visualize), how to pick the right visualization, and why results are only as good as the data.

What Programs Do with Data

- _____ — keep only rows that meet a condition
- _____ — arrange in order
- Combine / _____ data from multiple sources
- Compute statistics (sum, average, _____)
- _____ — charts and graphs

Tools for Insight

- Tables, diagrams, text, and visual tools communicate _____ from data.
- _____ tools help efficiently find information.
- Data _____ systems help find information and recognize patterns.
- _____ help organize data and find trends.

Filtering in Code

```
honor ← []
FOR EACH s IN scores
{
  IF (s ≥ 90)
  { APPEND(honor, s) }
}
```

If `scores` ← [95, 72, 90, 88], then `honor` = _____

Think:

Garbage in, garbage out! If the data are incomplete, biased, or messy, even a perfect program gives misleading results.



Key Vocabulary

Filtering

choosing a _____ of data that meets certain criteria

Sorting

arranging data into _____ (alphabetical, numerical, by date)

Data visualization

a _____ representation that reveals patterns

Transforming

changing every element in a data set in the _____ (e.g., °F → °C)

Iteration

repeating steps for _____ in a data set

You Try!

A movie table has columns: Title, Year, Genre, Rating (1–10), Box Office.
List the steps (filter / sort) to find the **highest-rated comedy released after 2015**.

Name: _____

Class: _____

Date: _____

Unit 2.4 Using Programs with Data

Go Deeper & Exam Practice

Work the Data

Name	Grade	Club	Hours
Ava	12	Robotics	14
Ben	11	Robotics	22
Cara	12	Drama	9
Dev	12	Robotics	8
Eli	12	Robotics	11
Fay	10	Drama	15

Filter Club = Robotics → _____ rows. Then Grade = 12
→ _____ rows. Then Hours > 10 → _____ rows.
Average hours of Drama members = _____.

Pick the Right Chart

- **Line graph** → change over _____
- **Bar graph** → compare _____
- **Pie chart** → parts of a _____
- **Scatter plot** → _____ between two variables

To show whether study time relates to test score, use a _____.

Exam Practice

★ Warm-up

1. Which is an example of **filtering** data?
- A. Converting prices from dollars to euros
 - B. Arranging products from cheapest to most expensive
 - C. Showing only products that cost under \$20
 - D. Adding up the price of every product

★★ On Level

2. Which visualization best shows how average global temperature changed from 1950 to 2020?
- A. Line graph
 - B. Pie chart
 - C. A single table cell
 - D. Word cloud

★★ On Level

3. A library database stores: title, author, genre, year published, and times checked out. Which steps find the **most-checked-out mystery book published after 2000**?
- A. Filter genre = mystery; filter year > 2000; sort by checkouts (high→low); take the first
 - B. Sort by checkouts; take the first; then filter genre = mystery
 - C. Filter year > 2000; sort by title; take the first
 - D. Filter genre = mystery; sort by year; take the last

★★★ Exam Hard

4. A data set has: student ID, grade level, number of absences, and GPA. Which **TWO** questions can be answered? (*Select two.*)
- A. Is there a relationship between absences and GPA?
 - B. Do students who live farther from school miss more days?
 - C. Which grade level has the highest average number of absences?
 - D. Why do students miss school?

Challenge!

A city uses step-count data from a fitness smartwatch app to decide where to build new parks. Identify a limitation of this data set and explain how it could lead to an unfair decision.

Unit 3.1 Variables and Assignments



The Big Picture

Variables are how programs remember things. You must trace assignment statements **exactly**, line by line — this is one of the most-tested skills on the AP exam.

Variables

- A variable is an _____ inside a program that can hold a value.
- Each variable holds _____ at a time (but that value can be a list).
- Use _____ names: `totalScore`, not `x`.
- Common data types: numbers, _____, strings, lists.

The Assignment Operator ←

`a ← expression`

- Evaluates the expression, then assigns a _____ of the result to `a`.
- The value stored is always the _____ value assigned.
- Read `a ← 5` as “a _____ 5.”
- `DISPLAY(a)` shows the value of `a`.

Key Vocabulary

Variable

a named _____ for a value that can change storing a value in a variable; _____ whatever was there before

Assignment

Data type

the kind of value (number, string, Boolean, list); determines what _____ make sense

Boolean

a value that is either _____

String

an ordered sequence of _____

The Swap Trap

To swap `a` and `b`, you need a _____ variable:

```
temp ← a
a ← b
b ← temp
```

Without `temp`, the original value of `a` is _____.

You Try!

Trace the code. What is displayed?

```
a ← 3
b ← a + 2
a ← b * 2
b ← a - b
c ← a + b
DISPLAY(a)
DISPLAY(b)
DISPLAY(c)
```



Think:

← is NOT “equals” from math. `x ← x + 1` means “take the current `x`, add 1, and store it back in `x`.”

Name:

Class:

Date:

Unit 3.1 Variables and Assignments

Go Deeper

Build a Trace Table

```
x ← 4
y ← x
x ← x + 6
y ← y * x
x ← y - x
```

Line	x	y
1	4	—
2	4	_____
3	_____	4
4	10	_____
5	_____	40

Copies, Not Links

```
x ← 7
y ← x
x ← 2
DISPLAY(y)
```

Displays _____. Line 2 gave `y` a _____ of `x`'s value. Changing `x` later does _____ change `y`.

`count ← count + 1` is used to _____.

Challenge!

Write code that **rotates** three variables: a gets b's value, b gets c's value, and c gets a's original value. Use one temporary variable and as few statements as possible. Then trace it with a = 1, b = 2, c = 3.

Unit 3.1 Variables and Assignments

Exam Practice

Exam Practice

★ Warm-up

1. What is displayed?

```
a ← 10  
b ← a  
a ← 20  
DISPLAY(b)
```

- A. b
- B. 20
- C. 30
- D. 10

★★ On Level

2. What are the values after this code runs?

```
p ← 10  
q ← 15  
r ← 20  
p ← q  
q ← r  
r ← p
```

- A. p = 15, q = 20, r = 10
- B. p = 20, q = 15, r = 10
- C. p = 10, q = 15, r = 20
- D. p = 15, q = 20, r = 15

★★ On Level

3. This code is meant to swap x and y:

```
x ← y  
y ← x
```

What does it actually do?

- A. Swaps x and y correctly
- B. Both variables end with the original value of x
- C. Both variables end with the original value of y
- D. Causes an error

★★★ Exam Hard

4. Which TWO code segments correctly swap the values of a and b? (Select two.)

```
I.  temp ← a    II.  a ← b  
    a ← b        b ← a  
    b ← temp  
  
III. a ← a + b   IV. temp ← a  
     b ← a - b   a ← b  
     a ← a - b   b ← a
```

- A. I
- B. II
- C. III
- D. IV

Show Your Work

Unit 3.2 Data Abstraction



The Big Picture

Lists manage complexity: instead of 100 separate variables, use ONE list. Know how AP lists work (**indexes start at 1!**) and why data abstraction makes programs easier to build and maintain.

Lists

- A list is an _____ sequence of elements.
- Each element has a unique _____ (position).
- On the AP exam, list indexes start at _____.
- `aList[i]` accesses the element at index `i`.
- Index less than 1 or greater than the length → _____; the program ends.

Why Data Abstraction?

- Separates the _____ of data from how it's actually stored.
- Gives a collection of data a single _____ without referencing each specific detail.
- Makes programs easier to _____.
- Code using a list can handle _____ of items.

Key Vocabulary

List

an ordered _____ of elements stored under one name

Element

an individual _____ in a list

Index

a common way to reference elements by their _____ in a list

Data abstraction

managing complexity by _____ a collection of data and hiding the details

Empty list

a list with _____, written `[]`

List Statements (Reference Sheet)

<code>aList ← [1, 2, 3]</code>	creates a list with _____ elements
<code>aList ← []</code>	creates an _____ list
<code>aList ← bList</code>	assigns a _____ of bList to aList



Think:

Python counts from 0, but the AP exam reference sheet counts from 1! `colors[1]` is the **FIRST** element.

You Try!

```
colors ← ["red", "blue", "green", "gold"]
DISPLAY(colors[2])
colors[3] ← colors[1]
DISPLAY(colors)
DISPLAY(colors[5])
```

What is displayed by each DISPLAY statement?

Name: _____

Class: _____

Date: _____

Unit 3.2 Data Abstraction

Go Deeper & Exam Practice

Without vs. With a List

```
score1 ← 88
score2 ← 92
score3 ← 75
avg ← (score1 + score2 + score3) / 3
```

```
scores ← [88, 92, 75]
avg ← sum(scores) / LENGTH(scores)
```

If a 4th score is added, version 1 needs a _____ and a new formula. Version 2 only needs _____ — this is the power of _____.

Index Practice

```
nums ← [10, 20, 30, 40, 50]
```

- `nums[LENGTH(nums)]` = _____
- `nums[2] + nums[4]` = _____
- `nums[nums[1] / 10]` = _____
- After `nums[1] ← nums[5] - nums[3]`, `nums[1]` = _____
- `nums[0]` → _____

Exam Practice

★ Warm-up

1. `myList ← [7, 3, 9]`. What is `myList[1]`?
- 3
 - 7
 - 9
 - Error

★★ On Level

2. What is a benefit of storing 30 test scores in a list instead of 30 separate variables?
- The scores can be processed with a loop, and the code works if the number of scores changes
 - Lists use no memory
 - Lists can only hold numbers, which prevents errors
 - Each score gets its own name

★★★ Exam Hard

3. Using the AP reference sheet, what does `a` contain after this code?

```
a ← [2, 4, 6, 8]
b ← a
b[2] ← 10
```

- [2, 4, 6, 8]
- [2, 10, 6, 8]
- [10, 4, 6, 8]
- [2, 4, 10, 8]

★★★ Exam Hard

4. Which expression ALWAYS gives the last element of a non-empty list `nums`?
- `nums[LENGTH(nums) - 1]`
 - `nums[LENGTH(nums)]`
 - `nums[0]`
 - `nums[LENGTH(nums) + 1]`

Challenge!

A phone stores contacts in two lists, `names` and `numbers`, where `names[i]` goes with `numbers[i]`. Explain what goes wrong if a name is removed from `names` but not from `numbers`. How would storing each contact as one combined record help?

Unit 3.3 Mathematical Expressions



The Big Picture

Every algorithm is built from **sequencing, selection, and iteration**. This topic covers sequencing and arithmetic — including **MOD**, which shows up on the exam ALL the time.

Building Blocks of Algorithms

- An algorithm is a _____ set of instructions that accomplishes a task.
- Every algorithm can be built using _____, _____, and _____.
- Sequencing = applying each step _____ the code statements are given.

Arithmetic Operators

+ - * / MOD

- $a \text{ MOD } b$ evaluates to the _____ when a is divided by b .
- $17 \text{ MOD } 5 = \underline{\hspace{2cm}}$
- MOD has the same precedence as _____.
- Order of operations: _____ first, then $*$ / MOD (left to right), then $+$ $-$.

Key Vocabulary

Algorithm

a finite set of _____ that accomplish a specific task

Sequencing

steps run _____ in order

Selection

choosing which steps to run based on a _____

Iteration

_____ steps

Expression

a value, variable, operator, or procedure call that is _____ to produce a single value

MOD Tricks

- $n \text{ MOD } 2 = 0 \rightarrow n$ is _____
- $n \text{ MOD } 10 \rightarrow$ the _____ of n
- If $a < b$, $a \text{ MOD } b = \underline{\hspace{2cm}}$ (ex: $3 \text{ MOD } 7 = 3$)
- $n \text{ MOD } k = 0 \rightarrow n$ is _____



Think:

MOD = "what's left over." $23 \text{ MOD } 4$: 4 goes into 23 five times (20) with 3 left over $\rightarrow 3$.

You Try!

Evaluate each expression.

(a) $25 \text{ MOD } 7$ (b) $6 \text{ MOD } 8$

(c) $3 + 12 / 4 * 2$ (d) $(3 + 12) \text{ MOD } 4 * 2$

(e) $40 \text{ MOD } 10$

Name:

Class:

Date:

Unit 3.3 Mathematical Expressions

Go Deeper & Exam Practice

Trace the Sequence

```
x ← 20
y ← x MOD 6
z ← x / y
x ← z MOD 4
DISPLAY(x + y + z)
```

y = _____ z = _____ x = _____

Displayed: _____

If lines 3 and 4 were swapped, would the output change? _____ – sequencing matters!

MOD in the Real World

- Clock: it's 10:00. What hour is it in 27 hours?
 $(10 + 27) \text{ MOD } 24 = \underline{\hspace{2cm}} \rightarrow 1:00 \text{ PM.}$
- Days: Day 0 = Sunday. 100 days later: $100 \text{ MOD } 7 = \underline{\hspace{2cm}} \rightarrow \underline{\hspace{2cm}}.$
- Every 3rd customer wins a prize: $\text{customerNum MOD } 3 = \underline{\hspace{2cm}}$

Exam Practice

★ Warm-up

1. What is the value of $13 \text{ MOD } 4$?

- A. 3
- B. 1
- C. 3.25
- D. 4

★★ On Level

2. Which expression evaluates to the ones digit of a positive integer n ?

- A. $n / 10$
- B. $10 \text{ MOD } n$
- C. $n \text{ MOD } 10$
- D. $n * 10$

★★ On Level

3. What is the value of $4 + 8 \text{ MOD } 3 * 2$?

- A. 6
- B. 8
- C. 12
- D. 2

★★★ Exam Hard

4. What is displayed?

```
a ← 7
b ← 3
a ← a MOD b + a
b ← a MOD (b + 2)
a ← (a + b) MOD 4
DISPLAY(a)
DISPLAY(b)
```

- A. 0 1
- B. 2 3
- C. 3 3
- D. 3 8

Challenge!

(a) Write a Boolean expression that is true when year y is a multiple of 4. (b) A game cycles through a list of 5 players. Write an assignment that moves `turn` from the current player to the next, wrapping from player 5 back to player 1.

Unit 3.4 Strings



The Big Picture

Strings are text. Know **concatenation** and **substrings**, and read the string procedures defined in each exam question very carefully. Watch for spaces — they count!

String Basics

- A string is an ordered sequence of _____.
- Strings are written in _____:
"hello"
- "5" is a string; 5 is a _____.
- Spaces and punctuation are _____ too.

Concatenation & Substrings

- Concatenation** joins two or more strings _____ to make a new string.
- `concat("Hi", "there")` → _____
(no automatic space!)
- A **substring** is _____ an existing string.

Key Vocabulary

String

an ordered sequence of _____

Concatenation

_____ strings together end-to-end

Substring

a _____ of an existing string

Empty string

a string with _____,
written ""

Length

the _____
in a string

Exam Procedures Are Defined for You

Questions define string procedures. Example:

`substring(str, start, length)` returns _____ characters of str beginning at position start. Positions start at _____.
`substring("computer", 1, 4)` → _____

Think:

Count spaces! "Hello" + " " + "World" has 11 characters. And a string of digits like "42" is text, not a number.

You Try!

word ← "PROGRAMMING" (positions start at 1)
(a) `substring(word, 1, 3)`
(b) `substring(word, 4, 4)`
(c) `concat(substring(word, 8, 4), substring(word, 1, 1))`

Name: _____ Class: _____ Date: _____

Class: _____ Date: _____

Date:

Go Deeper

Build a Username

```
first ← "Maya"
last ← "Lopez"
user ← concat(substring(last, 1, 4),
               substring(first, 1, 1))
user ← concat(user, "26")
```

After line 3: user = _____

Final: user = _____

The "26" is a _____, so it's joined, not added.

Number or String?

- `4 + 5` → _____
- `concat("4", "5")` → _____
- Number of characters in `"Hi there!"` → _____
- `concat("", "cat")` → _____
- `substring("banana", 3, 3)` → _____

Challenge!

Write ONE expression using `concat` and `substring` that builds a school email from variables `first` and `last`: first initial + full last name + `"@school.org"`. Example: `first = "Jamal", last = "Reed"` → `"JReed@school.org"`.

Unit 3.4 Strings

Exam Practice

Exam Practice

★ Warm-up

1. What does `concat("data", "base")` return?

- A. "data base"
- B. "basedata"
- C. "database"
- D. "data" "base"

★★ On Level

2. `substring(str, start, length)` returns *length* characters starting at position *start* (first position is 1). What does `substring("ALGORITHM", 3, 4)` return?

- A. "ORIT"
- B. "LGOR"
- C. "GORI"
- D. "GOR"

★★ On Level

3. How many characters are in the string "AP CSP 2027" ?

- A. 11
- B. 10
- C. 9
- D. 8

★★★ Exam Hard

4. `len(str)` returns the number of characters in *str*. What is displayed?

```
word ← "STAR"
result ← ""
i ← len(word)
REPEAT UNTIL (i < 1)
{
    result ← concat(result,
        substring(word, i, 1))
    i ← i - 1
}
DISPLAY(result)
```

- A. STAR
- B. RATS
- C. SSSS
- D. Error

Show Your Work

Unit 3.5 Boolean Expressions



The Big Picture

Boolean expressions evaluate to **true** or **false** and control every decision a program makes. Master relational operators, AND / OR / NOT, and equivalent expressions.

Relational Operators

= ≠ > < ≥ ≤

- Used to test the _____ between two values.
- `a = b` tests whether a and b are _____ (it does NOT assign).
- Every relational expression evaluates to a _____ value.

Logical Operators

- `NOT condition` → true if condition is _____
- `c1 AND c2` → true only if _____ are true
- `c1 OR c2` → true if _____ is true

Key Vocabulary

Boolean value

a value that is either _____

Boolean expression

an expression that _____ to true or false

Relational operator

compares two values (=, ≠, >, <, ≥, ≤) and produces a _____

Logical operator

_____ — combines or reverses Boolean values

Equivalent expressions

two expressions that produce the _____ for every input

Truth Table

A	B	A AND B	A OR B	NOT A
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Think:

Ranges use AND: `age ≥ 13 AND age ≤ 19` (a teen). Either/or situations use OR: `day = "Sat" OR day = "Sun"`.

You Try!

`x ← 8` and `y ← 3`. Evaluate:

- (a) `x > 5 AND y > 5`
 (b) `x > 5 OR y > 5`
 (c) `NOT (x = y)`
 (d) `(x MOD 2 = 0) AND NOT (y ≥ 3)`

Name: _____

Class: _____

Date: _____

Unit 3.5 Boolean Expressions

Go Deeper & Exam Practice

De Morgan's Laws (Exam Hard!)

- $\text{NOT } (A \text{ AND } B) \equiv (\text{NOT } A) \text{ ______ } (\text{NOT } B)$
- $\text{NOT } (A \text{ OR } B) \equiv (\text{NOT } A) \text{ ______ } (\text{NOT } B)$

Flip the operator AND flip each comparison:

$\text{NOT } (\text{age} \geq 13 \text{ AND } \text{age} \leq 19)$

$\equiv \text{age} \text{ ______ } 13 \text{ ______ } \text{age} \text{ ______ } 19$

The opposite of $>$ is ______ (not $<!$).

English $\rightarrow$ Boolean

- temp is between 60 and 80, inclusive:

- n is NOT divisible by 3:

- it's a weekend or a holiday:

- score is below 0 or above 100 (invalid):

Exam Practice

★ Warm-up

1. What does $(5 > 3) \text{ AND } (2 > 4)$ evaluate to?

- A. true
- B. Error
- C. 5
- D. false

★★ On Level

2. Which expression is true exactly when num is between 1 and 10, inclusive?

- A. $(\text{num} \geq 1) \text{ OR } (\text{num} \leq 10)$
- B. $\text{NOT } (\text{num} = 1)$
- C. $(\text{num} > 1) \text{ AND } (\text{num} < 10)$
- D. $(\text{num} \geq 1) \text{ AND } (\text{num} \leq 10)$

★★ On Level

3. Which expression is equivalent to $\text{NOT } (x > 10 \text{ OR } y < 5)$?

- A. $(x < 10) \text{ OR } (y > 5)$
- B. $(x < 10) \text{ AND } (y > 5)$
- C. $(x \leq 10) \text{ OR } (y \geq 5)$
- D. $(x \leq 10) \text{ AND } (y \geq 5)$

★★★ Exam Hard

4. A store gives a discount if a customer is a member AND spent at least \$50, OR if it's the customer's birthday. Which expression is correct?

- A. $\text{isMember AND } (\text{spent} \geq 50 \text{ OR } \text{isBirthday})$
- B. $(\text{isMember AND } \text{spent} \geq 50) \text{ OR } \text{isBirthday}$
- C. $\text{isMember OR } \text{spent} \geq 50 \text{ OR } \text{isBirthday}$
- D. $(\text{isMember OR } \text{spent} \geq 50) \text{ AND } \text{isBirthday}$

Challenge!

For which values of Boolean variables a and b is $(a \text{ AND } \text{NOT } b) \text{ OR } (\text{NOT } a \text{ AND } b)$ true? Fill in a truth table, then describe the pattern in one sentence.

a	b	$(a \text{ AND } \text{NOT } b) \text{ OR } (\text{NOT } a \text{ AND } b)$

Describe the pattern in one sentence: _____

Unit 3.6 Conditionals



The Big Picture

Conditionals (**selection**) let a program choose what to do. Trace IF and IF/ELSE carefully — exam questions love code where some branches never run, and the difference between one IF/ELSE and two separate IFs.

IF Statements

```
IF (condition)
{
  <block of statements>
}
```

- The block runs only if the condition is _____.
- If false, the block is _____ and the program continues after it.

IF / ELSE

```
IF (condition)
{ <first block> }
ELSE
{ <second block> }
```

- True → first block runs. False → the _____ block runs.
- Exactly _____ of the two blocks runs — never both, never neither.

Separate IFs vs. IF/ELSE

Two separate IF statements are _____ checked — both could run. An IF/ELSE always runs _____ branch.

On flowcharts, a condition is a _____ and an action is a _____.



Think:

Before tracing, ask: “Is this ONE IF/ELSE (only one branch runs) or several separate IFs (more than one could run)?”

Key Vocabulary

Selection

determines which parts of an algorithm run based on a condition being _____

Conditional statement

changes the sequential flow of control by running different statements based on a _____

Block

a group of statements inside _____

Flow of control

the _____ in which statements are executed

You Try!

```
score ← 85
bonus ← 0
IF (score > 80)
{ bonus ← bonus + 5 }
IF (score > 90)
{ bonus ← bonus + 10 }
ELSE
{ bonus ← bonus + 1 }
DISPLAY(bonus)
```

(a) What is displayed? (b) What if score were 95?

Unit 3.6 Conditionals

Go Deeper & Exam Practice

Flowchart → Code

Flowchart: $\diamond temp < 32?$ → **yes:** display "Freezing" · **no:** display "Not freezing"

```
IF ( _____ )
{ DISPLAY("Freezing") }
_____
{ DISPLAY("Not freezing") }
```

For temp = 32, it displays _____.

Spot the Bug

Should display "Pass" for grades of 70 or higher:

```
IF (grade > 70)
{ DISPLAY("Pass") }
ELSE
{ DISPLAY("Fail") }
```

It gives the wrong output when grade = _____.

Fix: change > to _____. This is a _____ error.

Exam Practice

★ Warm-up

1. What is displayed?

```
x ← 4
IF (x > 5)
{ DISPLAY("A") }
ELSE
{ DISPLAY("B") }
```

- A. A
- B. Nothing
- C. A B
- D. B

★★ On Level

2. What is the value of n after this code runs when n starts at 7?

```
IF (n MOD 2 = 0)
{ n ← n / 2 }
ELSE
{ n ← 3 * n + 1 }
```

- A. 3.5
- B. 7
- C. 21
- D. 22

★★★ Exam Hard

3. Which segments display "big" when $x > 10$ and "small" otherwise, for ALL integers x?

```
I. IF (x > 10) { DISPLAY("big") }
   ELSE { DISPLAY("small") }
II. IF (x > 10) { DISPLAY("big") }
    IF (x < 10) { DISPLAY("small") }
III. IF (x ≤ 10) { DISPLAY("small") }
     ELSE { DISPLAY("big") }
```

- A. I only
- B. II only
- C. I and III only
- D. I, II, and III

★★★ Exam Hard

4. What is displayed?

```
a ← 5
b ← 10
IF (a > b) { a ← b }
ELSE { b ← a }
IF (a = b) { a ← a + b }
DISPLAY(a)
DISPLAY(b)
```

- A. 10 5
- B. 5 10
- C. 10 10
- D. 15 10

Challenge!

Movie tickets cost \$8 for anyone under 12 or 65 and older, and \$12 otherwise. Write code using ONE IF/ELSE with a compound condition that sets price. Then list three test ages that would best check your code.

Unit 3.7 Nested Conditionals



The Big Picture

Nested conditionals are conditionals **inside** conditionals. They make multi-step decisions possible — and they're a favorite place for tricky exam questions about which branch runs.

What Is Nesting?

- Nested conditional statements consist of conditional statements _____ conditional statements.
- The inner IF is only checked if the outer condition leads _____.
- Nesting creates more than two possible _____ through the code.

Order Matters

In a chain of IF / ELSE IF checks, the _____ true condition wins and the rest are skipped.

If `score ≥ 70` is checked before `score ≥ 90`, a student with 95 gets a _____.

Rule: check the most _____ condition first.

Key Vocabulary

Nested conditional

a conditional statement placed _____ another conditional

Branch

one possible _____ through a conditional

Compound condition

a condition that combines Booleans with _____

Decision tree

a diagram that shows each _____ as a fork in the path

Equivalent code

different code that produces the _____ for all inputs

Nested IF vs. AND

```
IF (a)
{
  IF (b) { DISPLAY("X") }
}
```

is equivalent to `IF (a _____ b) { DISPLAY("X") }` — as long as there are no _____ branches.

You Try!

```
IF (temp > 80)
{
  IF (raining) { DISPLAY("Humid") }
  ELSE { DISPLAY("Beach") }
}
ELSE
{
  IF (temp < 40) { DISPLAY("Coat") }
  ELSE { DISPLAY("Jacket") }
}
```

temp	raining	Output
90	true	
90	false	
30	true	
80	false	



Think:

Draw it out! Sketch a decision tree — every IF is a fork in the road. Then follow one path for each test input.

Date:

Go Deeper

```
IF (score ≥ 90) { grade ← "A" }
ELSE
{
  IF (score ≥ _____) { grade ← "B" }
  ELSE
  {
    IF (score ≥ 70) { grade ← "C" }
    ELSE { grade ← _____ }
  }
}
}
```

```
IF (score ≥ 70) { grade ← "C" }
ELSE
{
    IF (score ≥ 80) { grade ← "B" }
    ELSE { grade ← "F" }
}
```

- A ride requires riders to be at least 48 inches tall. Riders under 54 inches must ride with an adult. Write a nested conditional using `height` that displays "Cannot ride", "Ride with adult", or "Ride alone". Then list the boundary heights you would test.

Unit 3.7 Nested Conditionals

Exam Practice

Exam Practice

★ Warm-up

1. What is displayed?

```
x ← 12
IF (x > 10)
{
  IF (x > 20) { DISPLAY("A") }
  ELSE { DISPLAY("B") }
}
ELSE { DISPLAY("C") }
```

- A. A
- B. B C
- C. C
- D. B

★★ On Level

2. Which is equivalent to the code below for all values?

```
IF (a > 0)
{
  IF (b > 0) { DISPLAY("yes") }
}
```

- A. IF (a > 0 OR b > 0) { DISPLAY("yes") }
- B. IF (b > 0) { DISPLAY("yes") }
- C. IF (NOT (a > 0)) { DISPLAY("yes") }
- D. IF (a > 0 AND b > 0) { DISPLAY("yes") }

★★ On Level

3. Which statement is true about this code?

```
cost ← 5
IF (weight > 10)
{ cost ← 20 }
ELSE
{
  IF (isMember) { cost ← 0 }
}
```

- A. A non-member with weight 4 pays 5
- B. A member with weight 12 pays 0
- C. A member with weight 10 pays 20
- D. Every member pays 0

★★★ Exam Hard

4. For which TWO inputs is "Go" displayed? (Select two.)

```
IF (NOT (light = "red"))
{
  IF (light = "green" OR clear)
  { DISPLAY("Go") }
}
```

- A. light = "red", clear = true
- B. light = "green", clear = false
- C. light = "yellow", clear = true
- D. light = "yellow", clear = false

Show Your Work

Unit 3.8 Iteration



The Big Picture

Iteration repeats code. Know **REPEAT n TIMES**, **REPEAT UNTIL**, and **FOR EACH**; be able to count how many times a loop runs, trace accumulators, and spot infinite loops.

REPEAT Loops

```
REPEAT n TIMES      REPEAT UNTIL
(condition)
{ <block> }         { <block> }
```

- **REPEAT n TIMES** runs the block exactly _____ times.
- **REPEAT UNTIL** runs until the condition is _____. The condition is checked _____ each pass.
- If the condition is already true at the start, the body runs _____ times.

FOR EACH & Infinite Loops

```
FOR EACH item IN aList
{ <block> }
```

- **item** is assigned each element of **aList** in order from _____.
- An infinite loop happens when the ending condition will _____ become true.

The Accumulator Pattern

```
sum ← 0
FOR EACH n IN nums
{ sum ← sum + n }
```

A sum must start at _____. A product must start at _____ (otherwise it's always 0!).



Think:

For **REPEAT UNTIL** ask: (1) Does the condition's variable change inside the loop? (2) Will it **EVER** make the condition true? If not → infinite loop!

Key Vocabulary

Iteration

a repeating portion of an algorithm; repeats a set number of times or until a _____ is met

Infinite loop

a loop whose ending condition _____ becomes true

Traversal

accessing _____ of a list, one at a time

Counter

a variable that tracks how many _____ something happens

Accumulator

a variable that _____ a total as the loop runs

You Try!

```
count ← 0
n ← 1
REPEAT UNTIL (n > 20)
{
  n ← n * 2
  count ← count + 1
}
DISPLAY(count)
DISPLAY(n)
```

What is displayed? Show the value of **n** after each pass.

Name: _____ Class: _____ Date: _____

Class: _____ Date: _____

Date:

Go Deeper

Count the Repeats

```
REPEAT 4 TIMES
{
    REPEAT 3 TIMES
    { DISPLAY("*") }
}
```

Stars displayed: _____

```
x ← 10
REPEAT UNTIL (x = 0)
{ x ← x - 3 }
```

x takes values 10, 7, 4, 1, _____ ... → this is an _____ . Fix: change the condition to _____ .

Trace a FOR EACH

```
nums ← [3, 8, 2, 7]
total ← 0
FOR EACH n IN nums
{
  IF (n > 4) { total ← total + n }
}
```

n	3	8	2	7
total	_____	_____	_____	_____

In words, this code

_____.

Challenge!

Write code that displays the first 10 multiples of 7 (7, 14, ..., 70) using **REPEAT UNTIL**. Then write a second version using **REPEAT n TIMES**.

Unit 3.8 Iteration

Exam Practice

Exam Practice

★ Warm-up

1. x starts at 1. What is x after this code?

```
REPEAT 5 TIMES  
{ x ← x + 2 }
```

- A. 10
- B. 11
- C. 7
- D. 12

★★ On Level

2. Which loop is an **infinite loop**?

- A. $i \leftarrow 1$, REPEAT UNTIL ($i = 10$) { $i \leftarrow i + 2$ }
- B. $i \leftarrow 1$, REPEAT UNTIL ($i > 10$) { $i \leftarrow i + 2$ }
- C. $i \leftarrow 10$, REPEAT UNTIL ($i < 0$) { $i \leftarrow i - 1$ }
- D. $i \leftarrow 0$, REPEAT UNTIL ($i \geq 5$) { $i \leftarrow i + 1$ }

★★ On Level

3. What is displayed?

```
result ← 0  
i ← 1  
REPEAT UNTIL (i > 10)  
{  
  IF (i MOD 2 = 0)  
  { result ← result + i }  
  i ← i + 1  
}  
DISPLAY(result)
```

- A. 25
- B. 20
- C. 55
- D. 30

★★★ Exam Hard

4. This code should compute the product of all numbers in nums, but always displays 0. Which change fixes it?

```
product ← 0  
FOR EACH n IN nums  
{ product ← product * n }  
DISPLAY(product)
```

- A. Change $\text{product} * n$ to $\text{product} + n$
- B. Move DISPLAY inside the loop
- C. Initialize product to 1 instead of 0
- D. Change FOR EACH to REPEAT UNTIL

Show Your Work

Unit 3.9 Developing Algorithms



The Big Picture

Different algorithms can solve the same problem. Know that algorithms can be built from scratch or by combining and modifying existing ones, how to tell if two algorithms are equivalent, and how to trace robot-on-a-grid problems.

Same Goal, Different Algorithms

- Algorithms can be written in different ways and still accomplish the _____.
- Algorithms that look similar can yield _____ results or side effects.
- Some conditional statements can be written as equivalent _____.
- Algorithms can be expressed in natural language, _____, and pseudocode.

Reuse Existing Algorithms

Using existing correct algorithms as building blocks reduces development _____, reduces testing, and simplifies finding errors. Common ones:

- find the maximum or _____
- compute a sum or _____
- check if a number is evenly _____ by another
- find a robot's path through a maze

Robot Commands

- `MOVE_FORWARD()` — one square in the direction it _____
- `ROTATE_LEFT()` — turns _____ counterclockwise
- `ROTATE_RIGHT()` — turns 90° clockwise
- `CAN_MOVE(dir)` — true if the square in that direction is _____ and on the grid
- Moving into a wall or off the grid → robot _____



Think:

To check if two algorithms are equivalent, test them with the **SAME** inputs — especially edge cases: 0, negatives, equal values, empty lists.

Key Vocabulary

Algorithm

a finite set of _____ that accomplish a task

Pseudocode

a plain-language _____ of code that isn't tied to one programming language

Flowchart

a _____ of an algorithm using shapes and arrows

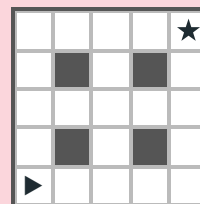
Equivalent algorithms

algorithms that give the _____ for every input

Building block

an existing, correct _____ reused to create a new one

You Try!



► = robot facing right
★ = goal
dark = wall

(a) Does this code reach the ★? Where does the robot end and which way does it face?

```
REPEAT 4 TIMES { MOVE_FORWARD() }
ROTATE_LEFT()
REPEAT 4 TIMES { MOVE_FORWARD() }
```

(b) Write a different algorithm that also reaches the ★.

Date:

Go Deeper

Unit 3.9 Developing Algorithms

Exam Practice

Exam Practice

★ Warm-up

1. Which is a benefit of building a new algorithm from existing, correct algorithms?
- A. It guarantees the new program has no errors
 - B. It makes the program run in constant time
 - C. It reduces development and testing time
 - D. It removes the need for documentation

★★ On Level

2. Algorithm I adds all the numbers and then divides by how many there are. Algorithm II divides each number by the count and then adds the results. Which is true?
- A. Only I computes the average
 - B. Both compute the average
 - C. Only II computes the average
 - D. Neither computes the average

★★ On Level

3. A robot faces **up**. Where does it end up?

```
REPEAT 3 TIMES
{
  MOVE_FORWARD()
  ROTATE_RIGHT()
}
```

- A. Back at the start, facing up
- B. One square right of the start, facing left
- C. One square above the start, facing right
- D. Three squares above the start, facing up

★★★ Exam Hard

4. Which TWO segments set `max` to the larger of `a` and `b` for ALL values, including when $a = b$? (*Select two.*)

```
I. IF (a > b) { max ← a }
   ELSE { max ← b }
II. max ← a
   IF (b > max) { max ← b }
III. IF (a > b) { max ← a }
     IF (b > a) { max ← b }
IV. max ← b
    IF (a < b) { max ← a }
```

- A. I
- B. II
- C. III
- D. IV

Show Your Work

Unit 3.10 Lists



The Big Picture

Lists are everywhere on the exam. Know every list procedure on the reference sheet — **INSERT, APPEND, REMOVE, LENGTH** — and how to traverse lists to search, filter, and compute.

List Procedures

- `INSERT(aList, i, value)`: shifts elements at index i and higher to the _____, puts value at index i .
- `APPEND(aList, value)`: adds value to the _____.
- `REMOVE(aList, i)`: removes the element at index i ; later elements shift _____.
- `LENGTH(aList)`: returns the number of _____.

Traversals

- A **complete** traversal visits _____ element.
- A **partial** traversal visits only _____ elements.
- Linear (sequential) search checks each element _____ until the value is found or all elements have been checked.

Common List Algorithms

- find the _____
- compute a sum or average
- _____ elements that match a condition
- filter matching elements into a _____



Think:

REMOVE inside a loop is dangerous: after every removal the list **SHIFTS** and gets shorter. Trace slowly and rewrite the list after each step!

Key Vocabulary

Traversal

visiting the elements of a list _____

Linear search

checks each element _____; works on any list

APPEND

adds a value to the _____ of a list

INSERT

adds a value at a specific _____ and shifts the rest right

REMOVE

deletes the element at an index; length _____

You Try!

```
pets ← ["cat", "dog", "fish"]
APPEND(pets, "bird")
INSERT(pets, 2, "frog")
REMOVE(pets, 4)
DISPLAY(pets)
DISPLAY(LENGTH(pets))
```

Write the list after each line.

Name: _____ Class: _____ Date: _____

Class: _____ Date: _____

Date:

Go Deeper

Linear Search Procedure

```
PROCEDURE contains(list, target)
{
    FOR EACH item IN list
    {
        IF (item = _____)
        { RETURN ( _____ ) }
    }
    RETURN ( _____ )
}
```

RETURN ends the procedure _____, so the search stops as soon as the target is found.

Worst case for a list of n items: _____ comparisons.

Removing in a Loop (Exam Hard!)

```

nums ← [4, 4, 7, 4]
i ← 1
REPEAT UNTIL (i > LENGTH(nums))
{
  IF (nums[i] = 4)
  { REMOVE(nums, i) }
  ELSE
  { i ← i + 1 }
}

```

Final nums: _____

Why is `i` only increased in the `ELSE`? So we don't _____ the element that shifts into position `i`.

Challenge!

Write a procedure `positives(nums)` that returns a NEW list containing only the values in `nums` that are greater than 0. Then show what `positives([3, -1, 0, 8, -5])` returns.

Unit 3.10 Lists

Exam Practice

Exam Practice

★ Warm-up

1. What does list contain after this code?

```
list ← [1, 2, 3]
APPEND(list, 4)
INSERT(list, 1, 0)
```

- A. [1, 0, 2, 3, 4]
- B. [1, 2, 3, 4, 0]
- C. [0, 2, 3, 4]
- D. [0, 1, 2, 3, 4]

★★ On Level

2. What does this code do?

```
count ← 0
FOR EACH w IN words
{
  IF (w = "yes")
  { count ← count + 1 }
}
```

- A. Counts how many elements equal "yes"
- B. Removes every "yes"
- C. Finds the index of the first "yes"
- D. Counts the words in the list

★★ On Level

3. This code is meant to find the minimum of nums.

```
min ← 0
FOR EACH n IN nums
{
  IF (n < min) { min ← n }
}
```

For which list does it work correctly?

- A. [5, 3, 9]
- B. [-2, -8, 4]
- C. [1, 2, 3]
- D. [7]

★★★ Exam Hard

4. What is result after this code?

```
list1 ← [10, 20, 30, 40]
result ← []
FOR EACH x IN list1
{
  IF (x MOD 20 = 0)
  { APPEND(result, x / 10) }
  ELSE
  { APPEND(result, x) }
}
```

- A. [1, 2, 3, 4]
- B. [2, 4]
- C. [10, 2, 30, 4]
- D. [10, 20, 30, 40]

Show Your Work

Unit 3.11 Binary Search



The Big Picture

Binary search is a MUCH faster way to find a value — but only if the data is **sorted**. Know the requirement, trace each step, count comparisons, and compare it with linear search.

How It Works

- Starts at the _____ of a sorted data set.
- Each step eliminates _____ of the remaining data.
- Repeats until the value is found or all elements are _____.
- Requirement: the data must be _____.

Linear vs. Binary

- Linear search works on _____ data.
- Binary search is often more _____ than linear search on sorted data.
- Worst-case checks for n items:
linear $\approx$ _____ binary $\approx$ _____ (+1)

Key Vocabulary

Binary search

repeatedly checks the middle and _____ of a sorted list

Linear search

checks elements _____ in order

Sorted

arranged in _____ (numeric or alphabetical)

Worst case

the input that requires the _____ steps
how much

Efficiency

_____ an algorithm uses as input grows

Worst-Case Checks

List size	Linear	Binary
8	8	_____
100	_____	_____
1,000	1,000	10
1,000,000	1,000,000	_____



Think:

Doubling the list size adds only **ONE** more step for binary search — but doubles the work for linear search.

You Try!

Use binary search on this sorted list. If the middle falls between two elements, use the one on the left.

[3, 8, 12, 15, 21, 27, 33, 40, 46]

(a) Search for **33**: list each value checked.

(b) Search for **5**: list each value checked and the result.

Unit 3.11 Binary Search

Go Deeper & Exam Practice

When CAN'T You Use It?

- If the list is unsorted, you must _____ or use linear search.
- Sorting takes time, so for a single search on unsorted data, _____ search may be the better choice.
- If the target is the first element, _____ search finds it in just 1 check.
- Binary search shines when a list is _____ and searched many times.

Count the Comparisons

A sorted list of 64 names:

- Binary search, worst case: _____ checks
($64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$)
- Linear search, worst case: _____ checks

The list grows to 128 names. Binary worst case: _____. Linear worst case: _____.

Exam Practice

★ Warm-up

1. What must be true about a list before binary search can be used?

- A. It must contain only numbers
- B. It must have an even number of elements
- C. It must be sorted
- D. It must have fewer than 1,000 elements

★★ On Level

2. About how many elements does binary search examine, in the worst case, in a sorted list of 500 items?

- A. 50
- B. 9
- C. 250
- D. 500

★★★ Exam Hard

3. Which TWO statements are true? (*Select two.*)

- A. Binary search can be used on any list.
- B. Linear search can be used on an unsorted list.
- C. For a large sorted list, binary search usually needs fewer comparisons than linear search.
- D. Binary search always finds a value faster than linear search.

★★★ Exam Hard

4. A sorted list grows from 1,000 items to 2,000 items. How does the worst-case number of binary search comparisons change?

- A. It doubles
- B. It increases by about 1
- C. It increases by 1,000
- D. It stays exactly the same

Challenge!

A music app has 10 million songs stored unsorted. A developer says, "Let's switch to binary search to make searching faster." Explain what else must happen first, and when this change would or wouldn't be worth it.

Unit 3.12 Calling Procedures



The Big Picture

Procedures package code so you can reuse it. You need to call procedures with arguments, trace what they return, follow the flow of control, and know **parameters** vs. **arguments**.

Procedures

- A procedure is a named group of programming instructions that may have _____ and return values.
- Also called methods or _____.
- Parameters are input _____ of a procedure.
- Arguments specify the _____ of the parameters when the procedure is called.

Calling & Returning

- A call interrupts sequential execution; the program runs the procedure's statements, then _____.
- Control returns to the point right _____ where the procedure was called.
- `RETURN(expression)` sends back the _____ of the expression and ends the procedure immediately.

Using a Returned Value

- `result ← procName(arg1, arg2)` stores the _____ in result.
- `DISPLAY(procName(5))` shows it.
- A procedure that only DISPLAYs does not _____ a value you can store!



Think:

Parameter = the placeholder in the DEFINITION. Argument = the actual value in the CALL.
 "Parameters are Placeholders, Arguments are Actual."

Key Vocabulary

Procedure

a _____ group of instructions that can be called

Parameter

a _____ variable in the procedure definition

Argument

the _____ passed in when calling

Return value

the value a procedure _____ to where it was called

Procedure call

a statement that _____ a procedure, e.g. `area(3, 7)`

You Try!

```
PROCEDURE mystery(a, b)
{
  IF (a > b) { RETURN (a - b) }
  RETURN (b * 2)
}
x ← mystery(10, 4)
y ← mystery(3, x)
DISPLAY(x + y)
```

(a) What is displayed? (b) Name the parameters and the arguments.

Date:

Go Deeper

Explain the difference between a procedure that **DISPLAYS** a result and one that **RETURNS** it. Give an example where only the RETURN version would work.

Unit 3.12 Calling Procedures

Exam Practice

Exam Practice

★ Warm-up

1. Given `PROCEDURE area(w, h)` and the call `area(3, 7)`, what are the arguments?

- A. w and h
- B. 3 and 7
- C. area
- D. 21

★★ On Level

2. What is the value of x?

```
PROCEDURE double(n)
{ RETURN (n * 2) }
x ← double(double(3))
```

- A. 12
- B. 9
- C. 6
- D. 36

★★ On Level

3. What is displayed?

```
PROCEDURE f(x)
{
  x ← x + 5
  RETURN (x * 2)
}
a ← 3
b ← f(a)
DISPLAY(a)
DISPLAY(b)
```

- A. 3 16
- B. 8 16
- C. 3 6
- D. 8 8

★★★ Exam Hard

4. What is displayed?

```
PROCEDURE firstOver(list, limit)
{
  FOR EACH item IN list
  {
    IF (item > limit) { RETURN (item) }
  }
  RETURN (-1)
}
DISPLAY(firstOver([4, 9, 2, 12], 5))
DISPLAY(firstOver([1, 2], 5))
```

- A. -1 -1
- B. 12 -1
- C. 9 12
- D. 9 -1

Show Your Work

Unit 3.13 Developing Procedures



The Big Picture

Writing your own procedures is **procedural abstraction** — and it's **REQUIRED** on the Create Performance Task. Know why procedures manage complexity and how parameters make them reusable.

Procedural Abstraction

- Provides a _____ for a process.
- Lets a procedure be used knowing only _____, not how it does it.
- Lets a large problem be solved using solutions to smaller _____.
- If a procedure's inside code changes but it still works the same, the rest of the program _____.

Modularity & Reuse

- Modularity = dividing a program into separate _____.
- Procedures let programmers _____ code and improve readability.
- Parameters let a procedure be _____ — reused with a range of input values.

```
PROCEDURE name(param1, param2)
{
  <instructions>
  RETURN (expression)
}
```

Key Vocabulary

Procedural abstraction

using a procedure by its name and behavior, _____ how it works

Modularity

breaking a program into separate _____

Generalization

using _____ so one procedure works for many inputs

Subproblem

a smaller piece of a _____

Duplicate code

copied code that should be replaced by a _____

Create Task Connection

Your Create PT program must include a student-developed procedure with:

- at least one _____ that affects its functionality
- an algorithm that includes _____, _____, and _____
- and the procedure must be _____ in your program.



Think:

If you're copy-pasting code and changing just one number, **STOP** — make a procedure and turn that number into a parameter!

You Try!

A program draws a square of size 10 and another of size 25 by copying the same code twice. Using `forward(n)` (move n steps) and `turnRight(90)`, write **ONE** procedure `drawSquare(size)`, then write the two calls.

Date:

Go Deeper

Design a procedure for a game that decides whether a player levels up. Give its **name**, **parameters**, what it **returns**, and write it in pseudocode. Then explain how it helps manage complexity.

Unit 3.13 Developing Procedures

Exam Practice

Exam Practice

★ Warm-up

1. Which best describes procedural abstraction?

- A. Writing every step of a program in one long block
- B. Removing all parameters from procedures
- C. Using a procedure knowing what it does without knowing how it does it
- D. Converting a program to binary

★★ On Level

2. Why might a programmer add a parameter to a procedure?

- A. To make the procedure run faster
- B. To prevent the procedure from returning a value
- C. So the procedure no longer needs to be called
- D. So one procedure can be reused with different input values

★★★ Exam Hard

3. Which procedure correctly returns how many times `target` appears in `list`?

```
I. count ← 0
   FOR EACH x IN list
   { IF (x = target) { count ← count + 1 } }
   RETURN (count)
II. FOR EACH x IN list
   { count ← 0
     IF (x = target) { count ← count + 1 } }
   RETURN (count)
III. count ← 0
    FOR EACH x IN list
    { IF (x = target) { RETURN (count + 1) } }
    RETURN (count)
```

- A. I only
- B. II only
- C. III only
- D. I and III

★★★ Exam Hard

4. A program calls `calculateTax(price)` in 20 places. A developer rewrites the inside of `calculateTax` to run faster, and it returns the same results as before. What else must change?

- A. All 20 calls must be rewritten
- B. Every procedure in the program must be rewritten
- C. The parameter must be renamed
- D. Nothing — the calls still work the same way

Show Your Work

Unit 3.14 Libraries



The Big Picture

You don't have to write everything from scratch! **Libraries** and **APIs** let you use code others wrote. Know what they are, how documentation helps, and why they simplify development.

Libraries

- A software library contains _____ that may be used in creating new programs.
- Existing code can come from internal or _____ sources, such as libraries or previously written code.
- Using libraries _____ the task of creating complex programs.

APIs

- An Application Program Interface (API) is a _____ for how the procedures in a library behave and can be used.
- _____ for an API/library is necessary to understand the behaviors it provides and how to _____ them.

Libraries You've Used

- Math library: `sqrt()`, `round()`
- Random: `RANDOM(a, b)`
- _____ library: drawing shapes
- Weather or maps _____: get live data from a service



Think:

Library = a toolbox of procedures.
API = the instruction manual that tells you each tool's name, what inputs it needs, and what it gives back.

Key Vocabulary

Library

a collection of _____ ready to be used in new programs

API

a _____ of how a library's procedures behave and are used

Documentation

explains each procedure's _____

Procedural abstraction

you can use a procedure without knowing _____ it works

Import

a statement that makes a library's procedures _____ in your program

You Try!

Documentation: `distance(lat1, lon1, lat2, lon2)` — *returns the distance in miles between two points.*

(a) Write a call that stores the distance from school (40.1, -75.2) to a museum (39.9, -75.1) in `d`.

(b) What did you need to know to use it? What did you NOT need to know?

Unit 3.14 Libraries

Go Deeper & Exam Practice

Read the Docs

```
translate(text, fromLang, toLang)
```

Returns the text translated into toLang. Language codes: "en", "es", "fr". Returns an error if a code is not valid.

- Number of parameters: _____
- What it returns: _____
- `translate("hola", "es", "en")` → likely _____
- `translate("hi", "en", "xx")` → _____

Pros & Cons

Pros: saves _____; code is often already _____; hides complexity.

Cons: you depend on code you don't control — it can _____ or contain bugs; may require Internet access; must follow licenses and give _____.

Exam Practice

★ Warm-up

1. What is an API?
- A. A type of computer virus
 - B. A specification for how procedures in a library behave and can be used
 - C. A programming language
 - D. A list of every bug in a program

★★ On Level

2. Which is a main benefit of using libraries?
- A. Programs never contain errors
 - B. They make programs use no memory
 - C. They simplify creating complex programs
 - D. They remove the need for testing

★★ On Level

3. A programmer wants to use a library procedure `sortList`. Which information is MOST necessary?
- A. The algorithm `sortList` uses internally
 - B. The name of the person who wrote it
 - C. What parameters it takes and what it returns or does
 - D. How many lines of code it contains

★★★ Exam Hard

4. Which TWO statements are true? (Select two.)
- A. Using library code requires understanding exactly how each procedure is implemented.
 - B. Libraries can reduce development time.
 - C. API documentation describes how to use a library's procedures.
 - D. Libraries can only contain procedures written by the same programmer.

Challenge!

A weather app relies on an external weather API. The API company releases an update that renames several procedures. Explain what happens to the app, what risk this shows, and one way developers can reduce that risk.

Unit 3.15 Random Values



The Big Picture

Random numbers power games, simulations, and security. Know exactly what **RANDOM(a, b)** can produce, find all possible outputs of code with randomness, and reason about probability.

RANDOM(a, b)

- Returns a random _____ from a to b, _____.
- Each value is _____ likely.
- `RANDOM(1, 6)` simulates a _____.
- Using random numbers means each run of a program may produce a _____ result.

Possible vs. Impossible

The exam often asks which outputs are POSSIBLE. List every value each `RANDOM` call can produce, then trace.

`RANDOM(1, 3) + RANDOM(1, 3)` ranges from _____ to _____. The sum _____ is most likely (it can be made 3 ways).

Key Vocabulary

Random number generator

produces values that can't be _____

Inclusive

the range _____ both endpoints

Probability

how _____ an outcome is, from 0 to 1

Outcome

one possible _____ of a random event

Pseudorandom

computer-generated numbers that _____ but come from an algorithm

Probability with RANDOM

- $P(\text{RANDOM}(1, 10) \leq 3) = \underline{\hspace{2cm}}$
- Something that happens 25% of the time: `IF (RANDOM(1, _____) = 1)`
- $P(\text{RANDOM}(1, 6) = 7) = \underline{\hspace{2cm}}$



Think:

`RANDOM(a, b)` has $(b - a + 1)$ possible values. `RANDOM(5, 10)` has 6 possible values — not 5!

You Try!

```
x ← RANDOM(1, 4)
y ← RANDOM(2, 3)
DISPLAY(x * y)
```

(a) Circle the possible outputs: 1 2 5 7 9 12

(b) List ALL possible outputs.

Unit 3.15 Random Values

Go Deeper & Exam Practice

Weighted Events

```
IF (RANDOM(1, 100) ≤ 30)
{ DISPLAY("rain") }
ELSE
{ DISPLAY("sun") }
```

P(rain) = _____ P(sun) = _____

Write a condition that is true with a 1-in-3 chance:

Random List Element

```
item ← list[RANDOM(1, _____)]
```

Common bug: `RANDOM(0, LENGTH(list))` can produce _____, which causes an _____ error, because AP lists start at 1.

For a list of 5 items, the chance of picking the first item is _____.

Exam Practice

★ Warm-up

1. Which value can `RANDOM(3, 8)` NOT produce?

- A. 3
- B. 9
- C. 8
- D. 5

★★ On Level

2. Which expression best simulates the sum of rolling two fair six-sided dice?

- A. `RANDOM(2, 12)`
- B. `RANDOM(1, 6) + RANDOM(1, 6)`
- C. `2 * RANDOM(1, 6)`
- D. `RANDOM(1, 12)`

★★ On Level

3. What is the probability that this code displays 3?

```
count ← 0
REPEAT 3 TIMES
{
  IF (RANDOM(1, 2) = 1)
  { count ← count + 1 }
}
DISPLAY(count)
```

- A. 1/2
- B. 1/8
- C. 1/6
- D. 1/3

★★★ Exam Hard

4. A spinner is red 50%, blue 25%, green 25%. Which code simulates it correctly?

```
I. n ← RANDOM(1, 4)
   IF (n ≤ 2) { c ← "red" }
   ELSE { IF (n = 3) { c ← "blue" }
         ELSE { c ← "green" } }
II. IF (RANDOM(1, 4) ≤ 2) { c ← "red" }
    ELSE { IF (RANDOM(1, 4) = 3) { c ← "blue" }
          ELSE { c ← "green" } }
```

- A. I only
- B. II only
- C. Both I and II
- D. Neither

Challenge!

Explain why calling `RANDOM` separately inside each `IF` of a chain (instead of storing one random value in a variable first) changes the probabilities. Use question 4 to support your answer.

Unit 3.16 Simulations



The Big Picture

Simulations are **abstractions** of real-world phenomena used to test hypotheses and make predictions — without the cost or danger of the real thing. Know their benefits, the simplifications they make, and their limits.

What Is a Simulation?

- Simulations are _____ of more complex objects or phenomena for a specific purpose.
- They use varying sets of values to reflect the changing _____ of a phenomenon.
- They often mimic real-world events to draw _____ without real-world constraints.

Simplifying & Bias

- Building a simulation involves _____ or simplifying functionality.
- Simulations can contain _____ from the choices of which real-world elements were included or excluded.
- Random number generators simulate the _____ that exists in the real world.

Why Simulate?

Use a simulation when a real experiment would be too:

- _____ or _____
- slow (evolution, climate) or fast (atoms)
- large (galaxies) or small (cells)

Simulations help form and refine _____.



Think:

Every simulation is a trade-off: more detail = more accurate, but slower and harder to build. Fewer details = faster, but it may miss important factors.

Key Vocabulary

Simulation

a program that _____ a real-world process or system

Hypothesis

a _____ that can be tested

Variability

natural _____ or variation in real-world events

Assumption

something the model treats as _____ to simplify reality

Abstraction

reducing complexity by _____ and keeping what matters

You Try!

A city wants to know if adding a left-turn lane will reduce traffic jams at an intersection.

(a) Why is a simulation a good choice? (b) Name two details that could be simplified. (c) Name one detail that should NOT be removed, and explain why.

Unit 3.16 Simulations

Go Deeper & Exam Practice

Coin Flip Simulation

```
heads ← 0
REPEAT 1000 TIMES
{
  IF (RANDOM(1, 2) = 1)
  { heads ← heads + 1 }
}
DISPLAY(heads)
```

Expected output: about _____. Will it always be exactly that? _____, because of randomness.

More trials → results get closer to the expected _____.

Limits of Simulations

- Results are only as good as the model's _____.
- A simulation can't perfectly _____ the real world.
- Leaving out a factor can introduce _____ into the results.
- Results should be checked against _____ when possible.

Exam Practice

★ Warm-up

1. Which is the BEST use of a simulation?
- Counting the students in a classroom
 - Sorting a list of names
 - Looking up a phone number
 - Testing how a new car design performs in a crash

★★ On Level

2. Which statement about simulations is true?
- They can test hypotheses without real-world constraints
 - They always produce exact predictions
 - They cannot use random numbers
 - They must include every real-world detail

★★ On Level

3. A flu simulation assumes every student has the same number of contacts per day. What is the most likely consequence?
- The results may not reflect real spread, because some students have many more contacts
 - The results will exactly match real life
 - The simulation will run forever
 - The simulation will not need random numbers

★★★ Exam Hard

4. Which TWO changes would most likely make a traffic simulation MORE accurate? (*Select two.*)
- Include rush-hour variation in the number of cars
 - Assume every driver drives exactly the speed limit
 - Use real traffic counts to set how often cars arrive
 - Remove all randomness so every run is identical

Challenge!

A biologist simulates a rabbit population: it starts at 100 and grows 20% each year. Write pseudocode that displays the population for each of 10 years. Then name one real-world factor missing from this model and how it would change the results.

Unit 3.17 Algorithmic Efficiency



The Big Picture

Some problems are solved quickly; others take “forever” as the input grows. Know how efficiency is measured, **reasonable vs. unreasonable** time, and when to use a **heuristic**.

Efficiency

- Efficiency estimates the computational _____ (time, memory) an algorithm uses.
- It's expressed as a function of the _____.
- It can be determined by formal reasoning, or informally by _____ the algorithm on different input sizes and measuring.

Reasonable vs. Unreasonable

- **Reasonable time:** polynomial efficiency or better — constant, _____, square (n^2), cube.
- **Unreasonable time:** _____ (2^n) or factorial ($n!$).
- Different correct algorithms for the same problem can have _____ efficiencies.

Key Vocabulary

Problem

a general description of a _____ that may or may not be solved with an algorithm

Instance

a problem with a specific _____

Decision problem

a problem with a _____ answer

Optimization problem

a problem that asks for the _____ solution

Heuristic

an approach that finds a solution that is _____, used when the exact solution is impractical

Watch It Grow

n	n^2	2^n
10	100	1,024
20	_____	_____
50	_____	$\approx 10^{15}$

Think:

Traveling salesperson: checking every route through 20 cities means $20! \approx 2.4$ quintillion routes. Instead, a heuristic like “always go to the nearest unvisited city” finds a good route fast.



You Try!

Classify each as **reasonable** or **unreasonable** time:

- (a) Find the max of a list (n steps) (b) Compare every pair of students (n^2 steps)
 (c) Try every subset of n items (2^n) (d) Check every ordering of n cities ($n!$)
 (e) Decision or optimization? “Is there a route under 100 miles?” / “What is the shortest route?”

Unit 3.17 Algorithmic Efficiency

Go Deeper & Exam Practice

Count the Steps

```
REPEAT n TIMES
{
  REPEAT n TIMES
  { DISPLAY("*") }
}
```

Steps: _____ $\rightarrow$ _____ time.

- A single loop over a list: _____ steps
- Binary search: about _____ steps
- Doubling n for an n^2 algorithm makes it about _____ times slower.

Heuristics

- GPS apps, chess programs, and delivery routes use heuristics because checking _____ would take too long.
- Using a heuristic trades a guaranteed _____ answer for _____.
- A heuristic is a good choice when an exact solution would take an _____ amount of time.

Exam Practice

★ Warm-up

1. An algorithm's number of steps grows as 2^n . Which best describes it?

- A. Constant time
- B. Linear time
- C. Unreasonable time
- D. Reasonable time

★★ On Level

2. Which is a **decision** problem?

- A. What is the shortest path from A to B?
- B. Is 97 a prime number?
- C. What is the largest value in a list?
- D. What is the best schedule for 30 classes?

★★ On Level

3. An algorithm takes 4 steps when $n = 2$, 9 steps when $n = 3$, and 16 steps when $n = 4$. Which best describes it?

- A. n steps — reasonable time
- B. 2^n steps — unreasonable time
- C. $n!$ steps — unreasonable time
- D. n^2 steps — reasonable time

★★★ Exam Hard

4. A delivery company must find the shortest route through 200 stops. Checking every route would take longer than the age of the universe. What is most appropriate?

- A. Buy a faster computer and check every route
- B. Choose a random route
- C. Give up; the problem is undecidable
- D. Use a heuristic that quickly finds a good, though not necessarily optimal, route

Challenge!

A program checks all subsets of n items (2^n steps). It takes 1 second when $n = 20$. Estimate how long it takes when $n = 30$ and $n = 40$. What does this show about exponential algorithms?

Unit 3.18 Undecidable Problems



The Big Picture

Some problems can't be solved by ANY algorithm — not just slowly, but at all. Know **decidable vs. undecidable** problems and the famous example: the **Halting Problem**.

Decidable vs. Undecidable

- A **decidable** problem is a decision problem for which an algorithm can be written to produce a _____ for all inputs.
- An **undecidable** problem is one for which _____ can be constructed that is always capable of providing a correct yes-or-no answer.

Some Instances CAN Be Solved

An undecidable problem may have some instances that have an algorithmic solution, but there is no algorithmic solution that solves _____ of the problem.

Example: we CAN tell that REPEAT 5 TIMES
 $\{ x \leftarrow x + 1 \}$ _____ — but no algorithm works for every program.

Key Vocabulary

Decision problem

a problem with a _____ answer

Decidable

an algorithm can solve _____ instance correctly

Undecidable

_____ algorithm can solve every instance correctly

Halting problem

deciding whether any program will _____ or run forever on a given input

Instance

the problem with one _____ input

The Halting Problem

In 1936, Alan _____ proved that no algorithm can decide, for every program and input, whether the program halts.

The proof uses a _____: assume such an algorithm exists, then build a program that does the opposite of what it predicts.

Think:

Undecidable $\neq$ slow! An unreasonable-time problem CAN be solved eventually. An undecidable problem can't be solved for all inputs — no matter how much time or computing power you have.



You Try!

Classify each as **decidable** or **undecidable**:

(a) Is this number even?

(b) Will this arbitrary program ever stop running?

(c) Is this list sorted?

(d) Is there a route through these 50 cities shorter than 1,000 miles? (Hint: is it impossible or just slow?)

Unit 3.18 Undecidable Problems

Go Deeper & Exam Practice

Proof Sketch (Contradiction)

1. Assume `halts(P, I)` exists and returns true if program P halts on input I.
2. Build `trouble(P) : IF`
`halts(P, P) → loop forever;`
`ELSE → stop.`
3. Run `trouble(trouble)`. If `halts` says it halts → it _____. If `halts` says it loops → it _____.
4. Either way, `halts` is _____, so it cannot _____.

Slow vs. Impossible

	Reasonable	Unreasonable	Undecidable
Example	sorting a list	_____	_____
Solvable for all inputs?	yes	_____	_____

Exam Practice

★ Warm-up

1. What does it mean for a problem to be **undecidable**?
 A. It takes a long time to solve
 B. It can only be solved by a heuristic
 C. It has no inputs
 D. No algorithm can always give a correct yes-or-no answer for all inputs

★★ On Level

2. Which statement is true?
 A. Some instances of an undecidable problem may be solvable by an algorithm
 B. An undecidable problem can be solved with a fast enough computer
 C. Every decision problem is decidable
 D. Undecidable problems have no correct answers

★★ On Level

3. Which best explains why the Halting Problem is undecidable?
 A. It has been proven that no algorithm can correctly decide halting for all program-input pairs
 B. Computers aren't fast enough yet
 C. Programs are too long to read
 D. It requires exponential time

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
 A. Every decision problem is decidable.
 B. An algorithm can determine whether certain specific programs halt.
 C. A problem that takes exponential time is undecidable.
 D. No algorithm can solve all instances of an undecidable problem.

Challenge!

A company claims its software can scan ANY program and guarantee it will never get stuck in an infinite loop. Use the Halting Problem to evaluate this claim. What could the software realistically do instead?

Unit 4.1 The Internet



The Big Picture

The Internet is a network of networks built on open protocols. Know how data travels in **packets**, what **protocols** do, and the vocabulary — this topic is vocab-heavy, so learn it well!

Networks

- A computing device is a physical artifact that can run a _____ (computers, phones, servers, routers, smart sensors).
- A computer network is a group of _____ computing devices that can send or receive data.
- A path is a sequence of directly connected devices from _____.
- Routing is the process of finding a _____ from sender to receiver.

Packets

- Information is passed through the Internet as a data _____.
- Data streams contain chunks of data in _____.
- Packets contain a chunk of data and _____ used for routing it.
- Packets may arrive in order, _____, or not at all.

Protocols to Know

- **IP** — gives each device an _____ and routes packets
- **TCP** — _____, in-order delivery; resends lost packets
- **UDP** — _____ but no guarantee of delivery or order
- **HTTP** — the protocol used by the _____



Think:

Internet = the roads. Web = one kind of traffic (web pages) on those roads. Email, games, and video calls use the Internet too, but they aren't the Web.

Key Vocabulary

Internet

a network of interconnected networks that use standardized, _____ (nonproprietary) protocols

Protocol

an agreed-upon set of _____ that specify the behavior of a system

Bandwidth

the maximum amount of data sent in a fixed amount of time, usually in _____

Router

a device that _____ packets toward their destination

World Wide Web

a system of linked pages, programs, and files that _____ the Internet

You Try!

(a) Why does the Internet use *open* protocols?
(b) Why might a video call use UDP instead of TCP?

(c) A friend says, "The Internet and the World Wide Web are the same thing." Correct them.

Unit 4.1 The Internet

Go Deeper & Exam Practice

Scalability & IP Addresses

- The Internet was designed to be _____ — able to grow in size to meet new demand.
- IPv4 uses _____ bits → about 4.3 billion addresses.
- IPv6 uses _____ bits → 2^{128} addresses.
- The switch was needed because the world was _____ of IPv4 addresses.

The Life of a Packet

1. A message is broken into _____.
2. Each packet gets metadata: source and destination _____, and a sequence number.
3. Routers forward packets, possibly along _____ paths.
4. At the destination, TCP puts them back in _____ and requests any _____ packets.

Exam Practice

★ Warm-up

1. What is a protocol?
- A. An agreed-upon set of rules that specify the behavior of a system
 - B. A type of computer hardware
 - C. A program that blocks viruses
 - D. The fastest path through a network

★★ On Level

2. Which best describes the relationship between the Internet and the World Wide Web?
- A. They are the same thing
 - B. The Internet is part of the Web
 - C. The Web is a system of linked pages and files that uses the Internet
 - D. The Web was created before the Internet

★★ On Level

3. Which TWO statements about packets are true? (*Select two.*)
- A. All packets from one message travel the same path.
 - B. Packets can arrive out of order.
 - C. Packets contain metadata used for routing.
 - D. Packets are only used to send email.

★★★ Exam Hard

4. Why has the Internet's use of open protocols been important to its growth?
- A. Any device or company can follow them to connect and communicate
 - B. Open protocols encrypt all data automatically
 - C. They make the Internet owned by one company
 - D. They prevent packets from being lost

Challenge!

Your home Internet has a bandwidth of 100 megabits per second. Ideally, how long does it take to download a 500 megabyte file? Show your work and explain why bits vs. bytes matters here.

Unit 4.2 Fault Tolerance



The Big Picture

The Internet keeps working even when parts of it break. Know **redundancy** and **fault tolerance**, and be able to read a network diagram to count paths and find weak spots.

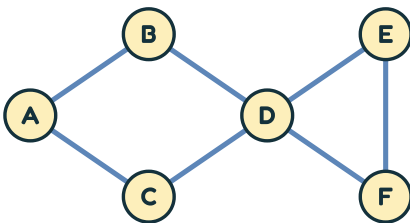
Redundancy

- The Internet has been engineered to be _____, with abstractions for routing and transmitting data.
- Redundancy is the inclusion of _____ that can be used to mitigate failure if other components fail.
- A system that can support failures and still continue to _____ is fault-tolerant.

Multiple Paths

- One way to get redundancy: more than one _____ between any two connected devices.
- If a device or connection fails, data is sent along a _____, if possible.
- Redundancy requires additional _____ but improves reliability.

Read the Network



- Number of different paths from A to F (no repeated devices): _____
- If device D fails, can A reach F? _____ → D is a single point of _____.



Think:

Read network diagrams like road maps: if one road closes, can you still get there another way? If not, you've found a weak spot.

Key Vocabulary

Fault tolerance

a system's ability to keep _____ when some parts fail

Redundancy

_____ components or paths used as backups

Single point of failure

one part whose failure _____ or stops the system

Path

a sequence of directly _____ devices from sender to receiver

Reliability

how _____ a system works correctly

You Try!

Use the network shown at left.

- List two different paths from A to F.
- If the connection D-E fails, can E still communicate with A? How?
- What is the minimum number of *connections* that must fail to stop A from reaching F? Which ones?

Unit 4.2 Fault Tolerance

Go Deeper & Exam Practice

Trade-offs

- Redundancy costs more _____ but increases reliability.
- Hospitals, banks, and cloud services use redundant servers and _____ because failure is costly.
- Adding a connection between _____ would mean no single device failure disconnects A from F.

Fault Tolerance Everywhere

- Backup _____ keep power on during an outage.
- Cloud services store copies of your data in _____ locations.
- Airplanes have redundant engines and _____ systems.
- TCP resends lost _____ — fault tolerance in software!

Exam Practice

★ Warm-up

1. What is redundancy in a network?
- A. Sending the same message to everyone
 - B. Compressing data
 - C. Removing unnecessary devices
 - D. Including extra components or paths that can be used if others fail

★★ On Level

2. Why is the Internet considered fault-tolerant?
- A. Its devices never fail
 - B. It uses only wireless connections
 - C. Every packet takes the same route
 - D. Data can be rerouted along other paths when a device or connection fails

★★★ Exam Hard

3. In the network shown on page 1, which connection, if added, would ensure that no single device failure could disconnect A from F?
- A. A-D
 - B. B-E
 - C. B-C
 - D. C-D

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
- A. Redundancy always lowers costs.
 - B. Fault-tolerant systems can keep working when some components fail.
 - C. Adding paths between devices can increase fault tolerance.
 - D. Fault-tolerant networks never lose packets.

Challenge!

A school network has only ONE connection to the Internet. Explain why this is not fault-tolerant, propose an improvement, and describe one trade-off of your improvement.

Unit 4.3 Parallel and Distributed Computing



The Big Picture

When one processor isn't fast enough, use more! Know **sequential**, **parallel**, and **distributed** computing, and be ready to CALCULATE run time and speedup — this shows up on the exam as math problems.

Three Models

- **Sequential:** operations performed _____.
- **Parallel:** a program is broken into smaller sequential operations, some performed _____.
- **Distributed:** multiple _____ are used to run a program.

Calculating Time

- A sequential solution takes as long as the _____ of all its steps.
- A parallel solution takes as long as its sequential tasks plus the _____ of its parallel tasks.
- **Speedup** = sequential time ÷ _____

Key Vocabulary

Sequential computing

steps are run _____, in order

Parallel computing

some steps run _____ on multiple processors

Distributed computing

a program runs across _____ (computers)

Speedup

sequential time _____ parallel time

Limits

- Parallel solutions scale more effectively than sequential ones.
- But speedup is eventually limited by the _____ portion that can't be split.
- Distributed computing solves problems too big for one computer because of processing time or _____ needs.



Think:

Parallel = more hands on the job. Four people folding laundry is much faster than one — but someone still has to wash it first, and that part can't be split!

You Try!

Three independent tasks take **40 s**, **30 s**, and **50 s**.

(a) Time on **ONE** processor?

(b) Fastest time on **TWO** processors? Show which tasks go where.

(c) What is the speedup?

Unit 4.3 Parallel and Distributed Computing

Go Deeper & Exam Practice

Dependencies

If task B needs the result of task A, they can't run in _____.

Tasks X (20 s) and Y (30 s) are independent. Task Z (10 s) needs results from both.

- Sequential time: _____ s
- Two processors: _____ + _____ = _____ s
- Speedup: _____

Distributed in Real Life

- Search engines split the web across thousands of _____.
- Folding@home lets volunteers' computers help study _____.
- Pros: solve problems too _____ for one computer.
- Cons: devices must _____ (overhead), plus security and coordination challenges.

Exam Practice

★ Warm-up

1. Which describes **parallel** computing?
- A. Operations run one at a time in order
 - B. A program runs only on a single core
 - C. Data is compressed before it's sent
 - D. A program is broken into smaller operations, some of which run at the same time

★★ On Level

2. Three independent tasks take 10 s, 15 s, and 20 s. On THREE processors, how long does the whole job take?
- A. 10 s
 - B. 20 s
 - C. 15 s
 - D. 45 s

★★ On Level

3. Four independent processes take 60, 30, 50, and 40 seconds. What is the minimum time on TWO processors?
- A. 60 s
 - B. 90 s
 - C. 100 s
 - D. 180 s

★★★ Exam Hard

4. A program takes 100 s on one processor. 20 s must run sequentially; the other 80 s can be split evenly among processors. How long on 4 processors?
- A. 25 s
 - B. 80 s
 - C. 20 s
 - D. 40 s

Challenge!

Using question 4: what is the shortest possible time with unlimited processors, and the maximum speedup? Explain why adding more processors eventually stops helping.

Unit 5.1 Beneficial and Harmful Effects



The Big Picture

Every computing innovation has effects — intended and unintended, helpful and harmful. On the exam and the Create task, you must identify both, and recognize that the same effect can help some people and hurt others.

Effects of Innovations

- People create computing innovations, and the way people complete tasks often _____ to use them.
- Not every effect of an innovation is _____ in advance.
- A single effect can be viewed as beneficial by some people and harmful by _____ — or even by the same person.
- Computing has increased _____ in other fields: medicine, engineering, communication, the arts.

Unintended Consequences

- Innovations can be used in ways their creators _____.
- The World Wide Web was first meant for rapid exchange of information within the _____ community.
- Machine learning and data mining enabled innovation in medicine, business, and science, but information discovered this way has also caused _____.

Rapid Sharing

Rapid sharing of a program, or running it with a large number of users, can result in significant impacts beyond the _____ or _____ of the programmer.



Think:

Use the “Who?” test: Who benefits? Who might be harmed? Who wasn't in the room when it was designed?

Key Vocabulary

Computing innovation

includes a _____ as an integral part of its function (physical, nonphysical software, or a concept)

Beneficial effect

an outcome that _____ people or society

Harmful effect

an outcome that _____ people, groups, or society

Unintended consequence

an effect that the creators _____ for

Responsible programming

considering the unintended _____ of a program before and after release

You Try!

Ride-share apps (like Uber and Lyft) connect riders with drivers.

(a) Give two beneficial effects. (b) Give two harmful effects. (c) Describe one effect that was probably unintended.

Name: _____

Class: _____

Date: _____

Unit 5.1 Beneficial and Harmful Effects

Go Deeper & Exam Practice

Same Effect, Different Views

- GPS location sharing: a parent sees _____; a teen may see a loss of _____.
- A social media feed that shows content you like: you enjoy _____, but it can trap you in a _____.
- Self-checkout machines: stores save money; cashiers may lose _____.

Analyze an Innovation

Innovation	Benefit	Harm
Facial recognition	_____	_____
Online learning	learn anywhere, anytime	_____
Your choice:	_____	_____

Exam Practice

★ Warm-up

1. Which is an example of a computing innovation being used in a way its creators did not originally intend?

- A. The World Wide Web, built for scientists, now used for shopping and social media
- B. A calculator used for addition
- C. A word processor used to type an essay
- D. A GPS used for directions

★★ On Level

2. Which statement is true about the effects of computing innovations?

- A. Every effect is predicted before release
- B. Innovations only have beneficial effects
- C. An effect can be beneficial to some people and harmful to others
- D. Harmful effects only come from bad programmers

★★ On Level

3. A fitness app publishes a public heat map of users' running routes. The map accidentally reveals the locations of secret military bases. Which best describes this?

- A. An intended beneficial effect
- B. A harmful, unintended consequence of a feature meant to be beneficial
- C. A syntax error
- D. A beneficial effect for everyone

★★★ Exam Hard

4. Which TWO actions are most responsible for developers? (*Select two.*)

- A. Considering how an app could be misused before release
- B. Assuming users will only use the app as intended
- C. Testing with diverse groups of users to discover potential harms
- D. Avoiding documentation to protect the code

Challenge!

Choose a computing innovation (it can be your Create task idea). Describe one beneficial effect and one harmful effect, and explain one specific way the harmful effect could be reduced.

Unit 5.2 Digital Divide



The Big Picture

Not everyone has equal access to computing. Know what the **digital divide** is, what factors cause it, who it affects, and how individuals, organizations, and governments can reduce it.

What Is It?

- Internet access varies between socioeconomic, geographic, and demographic characteristics, as well as between _____.
- The digital divide refers to differing _____ to computing devices and the Internet, based on socioeconomic, geographic, or _____ characteristics.
- It can affect both groups and _____.

Why It Matters

- The digital divide raises issues of equity, access, and _____, both globally and locally.
- It is affected by the actions of individuals, organizations, and _____.
- People without access can miss out on jobs, _____, health care, and government services.

It's More Than Owning a Device

The divide includes access to devices, reliable _____ Internet, and _____ (the skills to use technology). Rural areas may lack _____; cost is a barrier for many families.



Think:

During remote school, one student had fast Wi-Fi and a laptop; another shared a phone with siblings on a weak signal. Same assignment — very different experience.

Key Vocabulary

Digital divide

differing _____ to computing devices and the Internet

Socioeconomic

related to income, education, and _____

Geographic

related to _____ (rural vs. urban, country)

Demographic

related to characteristics like _____, race, or gender

Equity

_____ — everyone gets what they need to succeed

You Try!

A city moves ALL job applications online and closes its in-person office.

(a) How could this make the digital divide worse? (b) Propose two ways to reduce the impact.

Name: _____

Class: _____

Date: _____

Unit 5.2 Digital Divide

Go Deeper & Exam Practice

Name the Factor

- A rural town has no broadband service → _____
- A family can't afford a laptop or Internet bill → _____
- Many older adults are less comfortable using apps → _____
- A country's government blocks many websites → _____ action

Closing the Gap

- **Individuals:** donate devices, _____ others digital skills
- **Organizations:** schools lend laptops and _____; libraries offer free computers
- **Governments:** fund _____ in rural areas; subsidize Internet costs
- Smartphones help, but data caps and small screens limit tasks like _____.

Exam Practice

★ Warm-up

1. What is the digital divide?
- A. The difference between hardware and software
 - B. The gap between binary and decimal numbers
 - C. Differing access to computing devices and the Internet based on socioeconomic, geographic, or demographic characteristics
 - D. Splitting a program into modules

★★ On Level

2. Which action would MOST directly reduce the digital divide?
- A. Providing free Wi-Fi hotspots to low-income families
 - B. Releasing a more expensive phone
 - C. Requiring all forms to be completed online
 - D. Increasing the price of broadband

★★ On Level

3. A school requires all homework to be completed on a platform that needs high-speed video streaming. Which students are MOST likely disadvantaged?
- A. Students in rural areas with limited broadband
 - B. Students with new laptops
 - C. Students who type quickly
 - D. Students in after-school clubs

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
- A. The digital divide exists only between countries.
 - B. The digital divide can be affected by government actions.
 - C. The digital divide includes differences in access based on location.
 - D. Smartphones have completely eliminated the digital divide.

Challenge!

Some people argue that because most people own smartphones, the digital divide is basically solved. Evaluate this argument using at least two specific points.

Unit 5.3 Computing Bias



The Big Picture

Computers aren't automatically fair. Bias can be built into computing innovations through the **data** they learn from and the **choices** of the people who build them. Know where bias comes from and how to reduce it.

Where Bias Comes From

- Computing innovations can reflect existing human _____ because of biases written into the _____ or biases in the _____ used.
- Bias can be embedded at _____ of software development.
- Programmers should take action to _____ bias in algorithms as a way of combating existing human biases.

Real Examples

- Facial recognition is less accurate for people with darker skin when trained mostly on _____ faces.
- A hiring algorithm trained on past hires favored one _____ because past hiring did.
- Voice assistants misunderstand some _____ when the training data lacks them.

Reducing Bias

- Use diverse, _____ training data.
- Test with _____ users before release.
- Build diverse development teams (remember 1.1!).
- Collecting MORE data from the same biased source does _____ remove bias.



Think:

An algorithm trained on biased history will repeat that history — at computer speed and massive scale.

Key Vocabulary

Bias

unfair preference for or against a _____

Algorithmic bias

bias that results from an _____ design or its data

Training data

the data a machine learning model _____

Machine learning

computers learning patterns from _____ instead of explicit rules

Representative data

data that reflects the _____ it will be used on

You Try!

A bank uses an algorithm trained on 30 years of past loan decisions to approve new loans.

- (a) Explain how bias could enter this system.
(b) Suggest two ways to reduce it.

Unit 5.3 Computing Bias

Go Deeper & Exam Practice

Spot the Source

- A voice assistant struggles with accents because its recordings came from one region → bias in the _____.
- A programmer writes code so that “nurse” always defaults to female pronouns → bias in the _____.
- An app’s “normal” body temperature rule was set using only adult men → bias from a _____ sample.

Feedback Loops

A crime-prediction tool sends more police to neighborhoods with more past arrests.

More police → more _____ → the tool predicts even more crime there → even more police.

The algorithm’s _____ becomes its future _____, so the bias _____ over time.

Exam Practice

★ Warm-up

1. Which is MOST likely to lead to bias in a facial recognition system?
- A. Using a fast processor
 - B. Storing images in a compressed format
 - C. Training it mostly on images of one demographic group
 - D. Writing comments in the code

★★ On Level

2. What is the best way to reduce bias in a machine learning model?
- A. Collect more data from the same source
 - B. Remove all documentation
 - C. Train and test it on data that represents a diverse population
 - D. Run it on more computers

★★★ Exam Hard

3. A college admissions algorithm does not use race as an input, but it does use ZIP code. Why might it still be biased?
- A. ZIP codes are numbers
 - B. ZIP codes change every year
 - C. Algorithms can’t use location data
 - D. ZIP codes can be strongly correlated with race and income, acting as a proxy

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
- A. Bias can be introduced by the data used to train an algorithm.
 - B. Algorithms are objective, so they cannot be biased.
 - C. Programmers can take actions to reduce bias in algorithms.
 - D. Collecting more data always removes bias.

Challenge!

Connect Topic 1.1 (collaboration) to this topic: explain how a development team with diverse backgrounds and perspectives could reduce bias in a computing innovation. Give a specific example.

Unit 5.4 Crowdsourcing



The Big Picture

The Internet lets huge numbers of people contribute to one project. Know **crowdsourcing**, **citizen science**, and crowdfunding — and the benefits and challenges of collaborating at massive scale.

Crowdsourcing

- Crowdsourcing is the practice of obtaining input or information from a large number of people via the _____.
- Widespread access to information and public data helps people identify problems, develop _____, and share results.
- Crowdsourcing offers new models for collaboration, such as connecting businesses or social causes with _____.

Citizen Science

- Citizen science is scientific research conducted in whole or part by _____, many of whom may not be scientists.
- They contribute relevant data using their own _____.
- Examples: counting birds (eBird), classifying _____ (Galaxy Zoo), reporting weather.

Human Capabilities, Enhanced

Human capabilities can be enhanced by _____ via computing. Thousands of people each doing a small task can complete a _____ task — like Wikipedia or mapping traffic in real time (Waze).



Think:

Crowdsourcing works because many people each do a **SMALL** task. But quality control matters — anyone can contribute, including people who make mistakes or try to cause trouble.

Key Vocabulary

Crowdsourcing

getting input or information from _____ via the Internet

Citizen science

research done with help from _____ who contribute data

Crowdfunding

raising _____ from many people online

Public data

data made freely _____ for anyone to use

Collaboration

people working _____ toward a common goal

You Try!

A town wants to map every pothole on its roads.

(a) Describe how a crowdsourcing app could do this. (b) Give one benefit and one challenge of this approach.

Name: _____

Class: _____

Date: _____

Unit 5.4 Crowdsourcing

Go Deeper & Exam Practice

Pros & Cons

Pros: lots of data _____; low _____; many diverse ideas.

Cons: contributions may be _____ or malicious; results can be _____ toward people who have devices and free time (digital divide!); contributors' _____ must be protected.

Classify It

- Wikipedia → crowdsourced _____
- Kickstarter / GoFundMe → _____
- Volunteers count birds and upload sightings → _____
- Drivers report accidents on Waze → crowdsourced _____

Exam Practice

★ Warm-up

1. Which is an example of crowdsourcing?
- A. A company's 3 engineers design a new app
 - B. A computer running a sorting algorithm
 - C. A teacher grading tests alone
 - D. An online encyclopedia written and edited by thousands of volunteers

★★ On Level

2. Which is an example of citizen science?
- A. Volunteers record backyard bird sightings in an app that researchers analyze
 - B. A scientist writes a paper alone
 - C. A company sells science kits online
 - D. A student watches a science video

★★ On Level

3. A traffic app relies on drivers to report accidents. Which is a potential drawback?
- A. Drivers cannot use apps
 - B. The app cannot use GPS
 - C. Crowdsourcing always costs more than hiring staff
 - D. Reports may be inaccurate, and areas with few users get less coverage

★★★ Exam Hard

4. Which TWO are benefits of crowdsourcing? (*Select two.*)
- A. Many participants can solve problems too large for a small team.
 - B. It guarantees all data is accurate.
 - C. It can reduce the cost of collecting data.
 - D. It eliminates the need for any experts.

Challenge!

Explain how the digital divide (Topic 5.2) could affect the results of a crowdsourced project. Use the pothole app or another example.

Unit 5.5 Legal and Ethical Concerns



The Big Picture

Computing makes copying and sharing effortless — which raises legal and ethical questions. Know **intellectual property**, copyright, **Creative Commons**, open source, open access, and plagiarism.

Intellectual Property

- Material created on a computer is the _____ of the creator or an organization.
- Ease of access and distribution of digital information raises concerns about ownership, _____, and use.
- Measures should be taken to _____ intellectual property.
- Using computing to _____ individuals or groups raises legal and ethical concerns.

Ways to Share Legally

- **Creative Commons:** a public copyright license that enables the free distribution of an otherwise _____ work.
- **Open source:** programs made freely available that may be freely shared and _____.
- **Open access:** online research output free of all _____ on access and many restrictions on use.

Legal vs. Ethical

Legal = allowed by _____. **Ethical** = the _____ thing to do.

Using material created by someone else without permission and presenting it as one's own is _____ and may have legal consequences.



Think:

Ask two questions: "Is it allowed by law?" and "Is it the right thing to do?" Something can be legal but unethical — or the other way around.

Key Vocabulary

Intellectual property

creations of the _____
— code, art, music, writing
— owned by the creator

Copyright

legal protection that gives
creators _____ over
how their work is used

Creative Commons

a license creators use to
give others the right to
_____ their work

Open source

code that can be freely

Plagiarism

using someone else's
work without permission
and presenting it as

You Try!

A student finds a song on YouTube, uses it as background music in their game app without permission, and sells the app.

(a) What concern does this raise? **(b)** Give two legal alternatives.

Unit 5.5 Legal and Ethical Concerns

Go Deeper & Exam Practice

Creative Commons Licenses

- **BY** = you must give the creator _____
- **SA** (ShareAlike) = your new work must use the _____
- **NC** = _____ use only (no selling)
- **ND** = no _____ (no changes)

An image licensed **CC BY-NC** can be used in a free school project with _____, but not in an app you _____.

Ethics Case

A company trains an AI art generator on millions of artists' images without asking them.

- Is it legal? Still being _____ in courts.
- Ethical concerns: artists' _____, payment / compensation, and _____.
- Who might be harmed?

Exam Practice

★ Warm-up

1. What is open source software?
- A. Software that is always free of bugs
 - B. Programs made freely available that may be freely shared and modified
 - C. Software owned by the government
 - D. Software that can only be used offline

★★ On Level

2. A photographer wants anyone to use her photos as long as they give her credit. Which is the best option?
- A. Keep full copyright and allow no use
 - B. Delete the photos' metadata
 - C. Use a Creative Commons Attribution (BY) license
 - D. Use lossy compression

★★ On Level

3. A student uses code from an open-source library in their Create task. What is the most appropriate action?
- A. Use it and present it as their own original work
 - B. Avoid all outside code completely
 - C. Use it and acknowledge the source in their program and written responses
 - D. Rename the variables so it isn't recognized

★★★ Exam Hard

4. Which TWO statements are true? (*Select two.*)
- A. Material created on a computer is the intellectual property of its creator.
 - B. Anything posted on the Internet is free to use for any purpose.
 - C. Creative Commons licenses let creators allow sharing while keeping some rights.
 - D. Open access means anyone can claim authorship of the research.

Challenge!

Explain why something can be **legal but unethical**. Give a specific computing example and explain who could be harmed.

Unit 5.6 Safe Computing



The Big Picture

This is one of the most-tested topics in Big Idea 5! Know **PII**, authentication, **encryption** (symmetric vs. public key), and common attacks — phishing, malware, keylogging, and rogue access points.

PII & Privacy

- Personally identifiable information (PII) is information about an individual that _____, links, relates, or describes them (SSN, age, phone number, medical and financial info, biometrics).
- Search history, _____, and geolocation can be combined to build knowledge about a person.
- Information placed online can be used in unintended ways and is difficult to _____.

Authentication

- Authentication protects devices and information from _____ access.
- Strong passwords are easy to remember but hard to _____.
- **Multifactor authentication** requires at least _____ of: something you _____ (password), something you _____ (phone), something you _____ (fingerprint).

Attacks to Know

- **Phishing**: tricks a user into providing _____.
- **Keylogging**: records every _____ to steal passwords.
- **Virus**: malicious program that can _____.
- **Rogue access point**: unauthorized wireless network that can _____ data.



Think:

Public key = an open padlock you hand out to everyone. Anyone can snap it shut on a message — but only YOUR private key can open it.

Key Vocabulary

Encryption

the process of _____ data to prevent unauthorized access

Symmetric key

encryption that uses _____ for both encrypting and decrypting

Public key

pairs a public key for _____ with a private key for decryption

Certificate authority

issues digital certificates that _____ ownership of encryption keys (a trust model)

Malware

software intended to _____ a computing system or take partial control of it

You Try!

Name the threat:

(a) Email: “Your account is locked! Click here and enter your password.”

(b) A free game secretly records everything you type.

(c) At the airport, a hacker sets up a Wi-Fi network named “FreeAirportWiFi.”

(d) Explain why MFA protects you even if your password is stolen.

Unit 5.6 Safe Computing

Go Deeper & Exam Practice

Symmetric vs. Public Key

	Symmetric	Public key
Number of keys	1	_____
Same key locks & unlocks?	yes	_____
Main challenge	_____	slower

Caesar cipher (shift 3) is symmetric: CAT → _____

To send Bob a secret, Alice encrypts with _____ key. Only _____ key can decrypt it.

Protect Yourself

- Virus and malware _____ software helps protect against infection.
- Install _____ — they fix security vulnerabilities.
- Control app _____ (location, camera, contacts).
- Don't open unsolicited _____.
- Data sent over public networks can be intercepted, analyzed, and _____.
- Look for _____ — it means the connection is encrypted.

Exam Practice

★ Warm-up

1. Which is an example of **multifactor authentication**?

- A. A long password
- B. Two different passwords
- C. A password plus a code texted to your phone
- D. A security question

★★ On Level

2. Alice wants to send Bob a message using public key encryption. Which key should she use to encrypt it?

- A. Bob's public key
- B. Alice's private key
- C. Bob's private key
- D. Any symmetric key

★★ On Level

3. Which TWO are risks of sharing PII online? (*Select two.*)

- A. It can be combined with other data to build a profile of you.
- B. It is easy to permanently delete.
- C. It can be used for identity theft.
- D. Only the person who posted it can ever access it.

★★★ Exam Hard

4. What is the role of a certificate authority?

- A. It writes computer viruses
- B. It decrypts all Internet traffic
- C. It stores everyone's passwords
- D. It issues digital certificates that validate ownership of encryption keys

Challenge!

Explain how combining location data, cookies, and search history could reveal sensitive information about someone without their consent. Then recommend two specific steps they could take to protect their privacy.
